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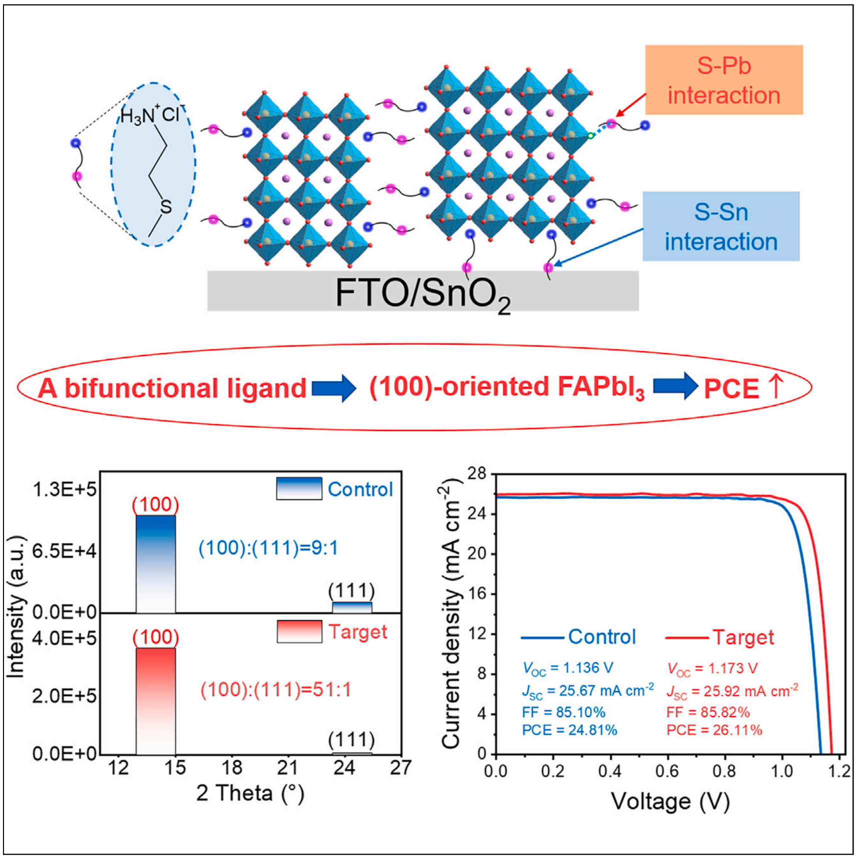

Perovskite solar cells (PSCs) have drawn significant attention due to their skyrocketed power conversion efficiency (PCE). Crystallization orientation and the buried interface have been proven to be key factors determining the efficiency of PSCs. Herein, we developed a bifunctional ligand 2-(methylthio) ethylamine hydrochloride (METEAM), concomitantly realized the (100)-oriented perovskite and well-contact buried interface. The resultant conventional-structure PSC devices delivered a PCE of 26.1% (certified 25.8%) with improved operational and ambient stability.

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Highlights

A bifunctional ligand METEAM induces (100)-oriented perovskite crystallization

METEAM molecules spontaneously aggregate at the buried interface

METEAM operates as a bridge between perovskite and ETL

A high PCE of 26.11% (certified: 25.80%) is achieved

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Bifunctional ligand-induced preferred crystal orientation enables highly efficient perovskite solar cells

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SUMMARY

Crystallization orientation and the buried interface have been proven to be key factors determining the efficiency of perovskite solar cells (PSCs). Here, we report a facile strategy to concomitantly induce (100)-oriented perovskite and improve buried interface by incorporating a bifunctional ligand 2-(methylthio) ethylamine hydrochloride (METEAM) into perovskite precursor solution. METEAM molecules preferentially adsorb on (100) facets of perovskite via strong interactions with perovskite lattice to induce oriented perovskite crystallization. Meanwhile, METEAM molecules spontaneously aggregate at the buried interface and operate as a bridge between the perovskite and tin oxide (SnO_2) electron transport layer to bidirectionally passivate their defects. As-prepared perovskite films exhibit suitable energy level and high mobility for interfacial charge transfer, low trap state density, and long carrier lifetime. The resultant conventional-structure PSC devices deliver a power conversion efficiency (PCE) of 26.1% (certified 25.8%) with improved operational and ambient stabilities, which is among the highest PCE of conventional PSCs.

CONTEXT & SCALE

The high quality of perovskite light-absorbing layer and reduced interfacial defects have brought a significant improvement to the power conversion efficiency (PCE) of perovskite solar cells (PSCs). It is well acknowledged that regulating perovskites crystal nucleation and growth by incorporating additive is a facile and effective strategy, while concomitantly improving the buried interface is extremely difficult without a pre-spin-coating passivator. To further boost PCE of PSCs, it is of vital significance to develop a strategy to simultaneously address the above-mentioned key issues. Herein, a bifunctional ligand 2-(methylthio) ethylamine hydrochloride (METEAM) was incorporated into the perovskite precursor solution, inducing concomitantly the (100)-oriented perovskite and well-contact buried interface, and the resultant conventional-structure PSC devices deliver a PCE of 26.1% (certified 25.8%), which is among the highest PCE of conventional PSCs.

INTRODUCTION

Metal halide perovskite is regarded as the ideal candidate for photovoltaic applications due to its unique optoelectronic properties, including high absorption coefficient, tunable band gap, high carrier mobility, long carrier diffusion length, and lifetime.[1–4] Perovskite solar cells (PSCs) utilizing metal halide perovskites as light absorbers have garnered tremendous attention over the past decade, with the power conversion efficiency (PCE) of the state-of-the-art single-junction PSCs exceeding 26.0%,[5] making them promising candidate for commercial deployment. To promote commercialization, the exploration of effective and simplified strategies to further boost the PCE of PSCs becomes significant. In the last several years, great attention has been paid to the development of defects passivation materials and charge transporting materials to suppress the non-radiative recombination and facilitate the charge transfer, contributing to the significantly enhanced PCE.[6–13] As these techniques become mature, it requires other aspects to be considered to foster the efficiency to approach the Shockley-Queisser (S-Q) limit.[14]

One critical factor that determines the efficiency and stability of PSCs has been proven to be crystal orientation.[6,15–19] Different crystal facets of perovskite possess distinct chemical, physical, and photoelectric characteristics.[16–20] For example, previous studies unveiled that photocurrents of these facets were in the order of (100) > (111) > (110). The well-defined (100) and (111) facets exhibited more n-type characteristics and higher carrier mobility than those of the (110) facets.[16] Theoretically, the (100) and the (111) facets have high dielectric constant due to the different density and symmetry of atoms, leading to low exciton binding energies, which is beneficial for generating free carriers.[17] Dependence of trap density on the surface crystallographic facets using density-functional theory (DFT) calculations based on a cubic $FAPbI_3$ model indicates there are no trap states that appear for the (100) surface in either formamidinium iodide (FAI)-rich or PbI_2-rich termination cases; however, (110) and (111) surfaces would introduce additional trap states.[18] In addition, the facet-dependent degradation study indicates the α- to δ-phase transformation is accelerated selectively on the (100) facets by moisture, whereas phase transformation is considerably retarded on the (111) facets because of its thermodynamic stability against surface interaction with water molecules.[19] Particularly, for the most common $FAPbI_3$ perovskites applied in high-efficiency devices, the realization of preferred orientation via changing the surface energies and reaction parameters could greatly impact the optoelectrical properties, with high PCE over 25% achieved recently.[16,18,20–24] Organic ligands containing $–H_3N^+$ "head" is an ideal candidate for interacting with the perovskite framework via incorporating the head into the A site of the perovskite lattice. Another key factor that has been previously overlooked is the quality of the buried interface, which could affect the undesirable non-radiative recombination losses at the bottom interface.[25–28] Sulfur (S)-containing ligand is an effective passivator for uncoordinated metal atoms within tin oxide (SnO_2) and perovskite, resulting in improved electron transport layer (ETL) and active layer.[29,30] Although some efforts have been made to address this issue in the last 2 years, those strategies mainly rely on the embedding of an additional interfacial layer, adding to the complexity of device fabrication.[25,31–33] Considering these two aspects, developing facile strategies to simultaneously regulate the crystallinity orientation and buried interface becomes crucial to further boost the efficiency of PSCs toward commercialization.

Herein, we report a facile and effective strategy to concomitantly induce the highly oriented $FAPbI_3$ perovskite and improved buried interface by incorporating a bifunctional ligand 2-(methylthio) ethylamine hydrochloride (METEAM), a chloride containing $–H_3N^+$ head and S-containing "tail," into perovskite precursor solution. METEAM molecules form a gradient vertical distribution across perovskite film, spontaneously aggregating at the buried interface, thus operating as a bridge between the perovskite and SnO_2 ETL and bidirectionally passivating their defects. Besides, METEAM molecules preferentially adsorb on (100) facets of perovskite via their strong interactions with perovskite lattice, thereby decreasing the surface energy and inducing (100)-oriented perovskite crystallization. Meanwhile, the phase transition temperature of α-phase $FAPbI_3$ is reduced, promoting the transformation to photovoltaic black-phase α-$FAPbI_3$. As-prepared $FAPbI_3$ perovskite films exhibit low trap state density, long carrier lifetime, and high carrier mobility. As a result, as-prepared conventional-structure PSC devices deliver a PCE of 26.1% (certified 25.8%) with improved operational stability, which is among the highest PCE of conventional PSCs.

RESULTS AND DISCUSSION

Regulating buried interface and crystal orientation by METEAM

The $FAPbI_3$ perovskite films were deposited via a one-step process with a small amount of METEAM (0.6 mol %) incorporated into the precursor solution to induce ideal crystal orientation and a well-contact buried interface (Figure 1A). The

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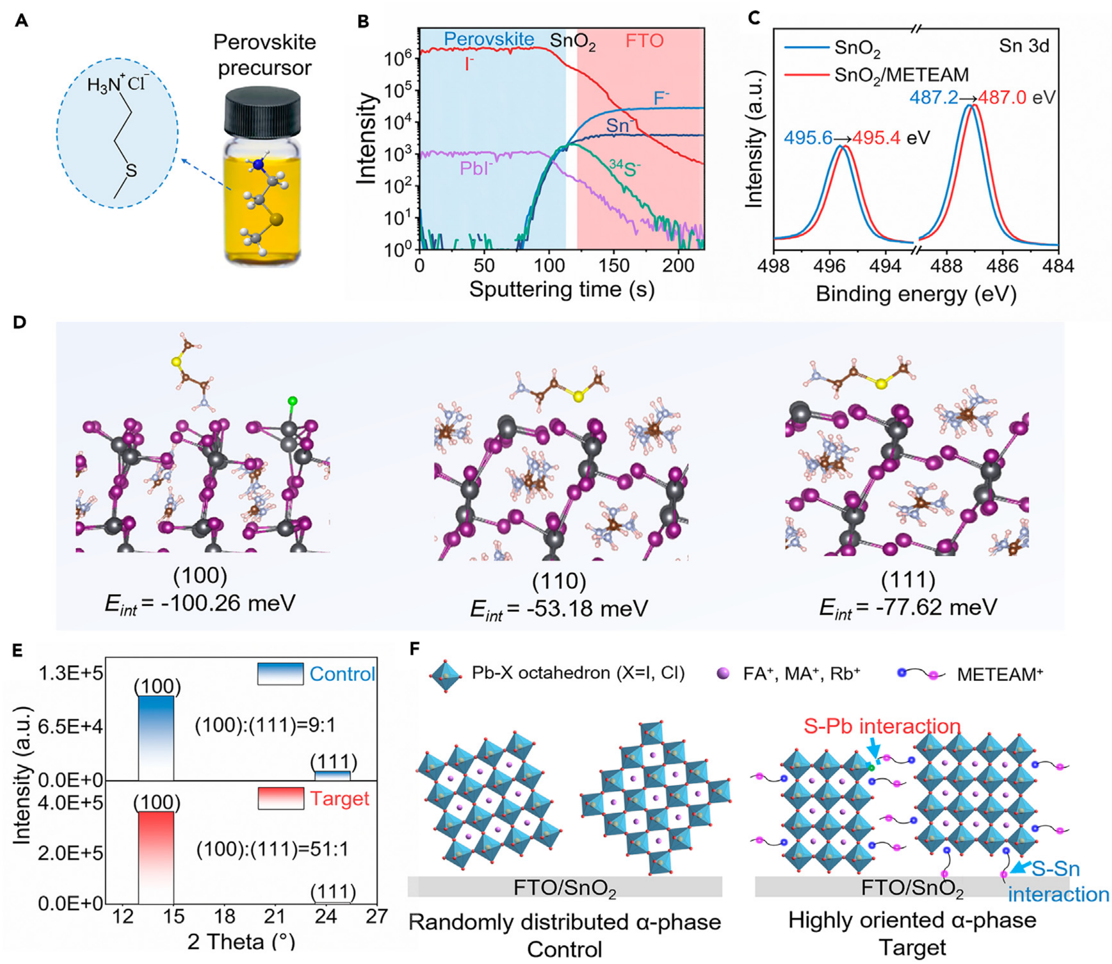

Figure 1. The formation of (100)-orientation $FAPbI_3$ and self-aggregation of METEAM molecules at the buried interface

(A) The molecular structure of METEAM and the schematic diagram of $FAPbI_3$ perovskite precursor solution with METEAM incorporation.

(B) ToF-SIMS of the target perovskite film deposited on the glass/FTO/SnO_2 substrate.

(C) Sn 3d high-resolution XPS profiles of SnO_2 with and without METEAM.

(D) Equilibrium configuration of METEAM-α-$FAPbI_3$ with (100), (110), and (111) facets for the fastest formation with the interaction energies (E_{int}). Detailed DFT calculation data are shown in Table S2.

(E) Peak intensities of (100) and (111) facets in the $FAPbI_3$ perovskite films with and without METEAM obtained from X-ray diffraction (XRD) pattern (Figure 2J). The intensity of (110) facets in the XRD profile is weak and thus omitted.

(F) Schematic diagram of the crystal growth process of the $FAPbI_3$ with and without METEAM.

corresponding $FAPbI_3$ perovskites fabricated without and with METEAM incorporation are denoted as "control" and "target," respectively. To investigate the vertical distribution of METEAM in perovskite film, depth profiling using time-of-flight secondary ion mass spectrometry (ToF-SIMS) was performed. As shown in Figure 1B, a pronounced S signal from METEAM was observed at the perovskite/SnO_2 buried interface, which indicates the aggregation of METEAM molecules at the bottom interface during perovskite film formation. It is noteworthy that the S signal of samples grown on the substrate without the SnO_2 ETL exhibits a similar distribution, counteracting the influence of S elements used in the chemical bath deposition (CBD) preparation of SnO_2 (Figure S1).

To investigate the underlying mechanism behind the spontaneous aggregation of METEAM ligands at the buried interface, dynamic light scattering (DLS) measurements are carried out. As seen from Figure S2, in addition to the observation of the colloidal particles with sizes less than 10 nm in both precursor solutions, after adding 0.6 mol % METEAM, clearly the proportion of the colloids with large sizes in range of 1–6 μm increases in the target precursor solution, along with decreased proportion of small-sized colloids. We propose that the target precursor solution composed of METEAM tends to form larger iodine-shared, large-sized lead polyiodide colloids due to the adsorption of METEAM molecules.[34] During the one-step solution-processed perovskite film deposition, the top precursor solution becomes supersaturated firstly along with precipitation of the intermediate phase as the anti-solvent dripped.[35] Since $METEAM^+$ is a larger spacer cation than FA^+ and MA^+, its incorporation into the top-surface intermediate phase composed likely of FA-, MA-, and Rb-based 3D perovskite becomes more difficult owing to its larger formation energy for forming 2D perovskite with large *n* values.[36] Consequently, as the supersaturated region moves from the top to the bottom during the crystallization process, the large-sized colloids adsorbed with METEAM gradually approach the bottom, which is the buried interface between SnO_2 and perovskite, leading to a vertical gradient of the METEAM ligand.

Moreover, X-ray photoelectron spectroscopy (XPS) measurement was conducted to investigate the interfacial interaction. After incorporating METEAM, a shift of Sn 3d core level of SnO_2 to lower binding energies by $\sim$0.2 eV along with the shift of S 2p core level to a higher level in METEAM indicates strong interactions between METEAM and SnO_2 interface (Figures 1C and S3A–S3D),[29] which provide a driving force for crystal deflection and induce oriented crystal growth. Meanwhile, the contents of hydroxyl oxygen (O_{OH}) and oxygen vacancies (O_v) estimated from O 1s XPS profile decrease after incorporating METEAM (Figures S3E and S3F; Table S1), indicating that the trap states such as surface hydroxyl groups and oxygen vacancies on the SnO_2 surface are effectively passivated.[28,37] In addition, as inferred from the negative shifts of the Pb 4f XPS signals of $FAPbI_3$ perovskite, the underlying $METEAM^+$ may additionally render S-Pb coordination interactions and consequently passivate the defect-enriched perovskite/SnO_2 buried interface Figure S4.[30] Hence, METEAM behaves as a bifunctional ligand bidirectionally passivating the defects of both SnO_2 and perovskite.

DFT calculations were carried out to investigate how METEAM molecules interact with the Pb-I framework of $FAPbI_3$ perovskite.[16] Equilibrium configurations of METEAM-α-$FAPbI_3$ with (100), (111), and (110) facets for the fastest formation with the interaction energies are given in Figure 1D and Table S2. The $METEAM^+$ exhibits the highest interaction energy (E_{int}) of -100.26 meV with (100) facets and depicts lower E_{int} values of -53.18 and -77.62 meV with (110) and (111) facets, respectively. We also calculated E_{int} for the $METEAM^+$ with –S–CH_3 group and –H_3N^+ group exposed to the (100) facet of $FAPbI_3$ perovskite, which are -91.19 and -100.26 meV, respectively. The larger value of the latter suggests a spontaneous transition from the former to the latter configuration (Figure S5). This suggests that the $METEAM^+$ preferentially adsorbs on the (100) facet of perovskite via the –H_3N^+ group to decrease its surface energy (Figure 1D), which may act as the thermodynamic driving force to induce the preferred (100)-orientation of the black-phase α-$FAPbI_3$ perovskite.[16,23] Furthermore, Figure 1E compares the peak intensities of (100) and (111) facets extracted from the X-ray diffraction (XRD) patterns of the control and target perovskite films. The growth of crystal planes in the control film shows random orientation with relatively small relative peak intensity of (100):(111) crystal plane (9:1), whereas the target film exhibits a considerable enhancement in the relative peak intensity of (100):(111) crystal plane (51:1), indicating that METEAM incorporation promotes (100)-oriented crystallization of perovskite film.

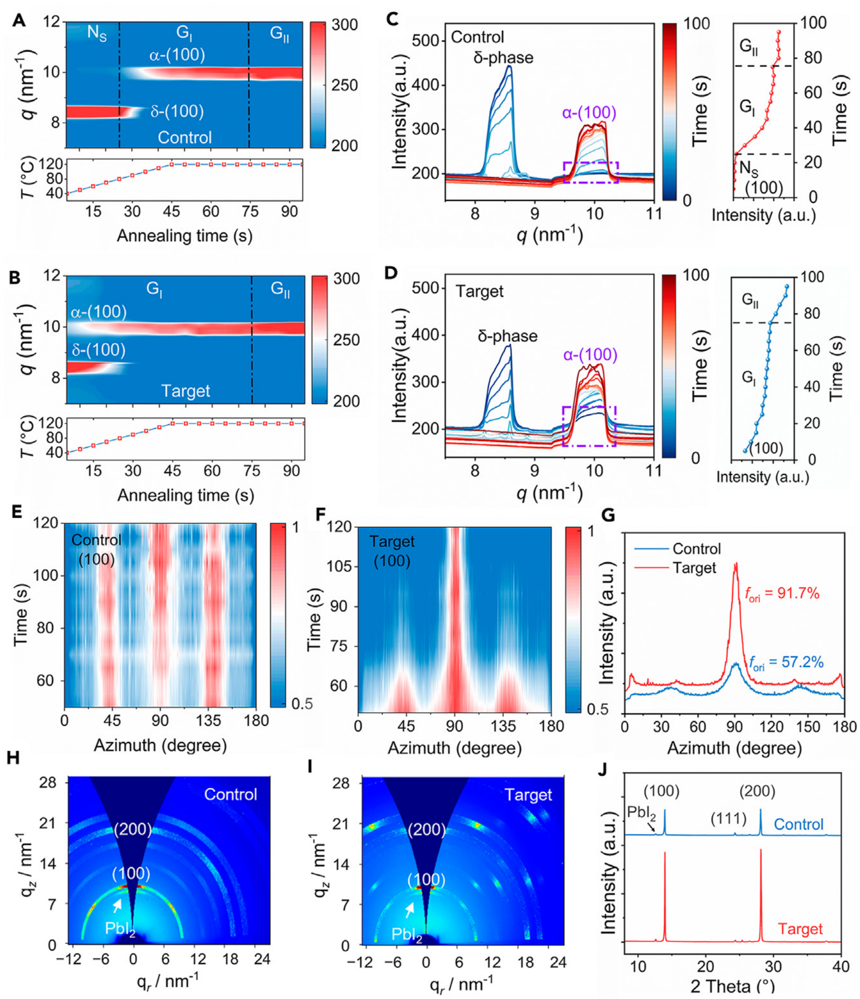

Figure 2. *In situ* GIXRD monitoring of the crystallization process for perovskite films

(A and B) *In situ* GIXRD analyses of perovskite films fabricated without (A) and with (B) METEAM, where N_s denotes the nucleation stage, G_I and G_{II} denotes the growth stage.

(C and D) The GIXRD intensity profiles along the q_z direction and the (100) plane intensity extracted from the *in situ* thermal annealing GIXRD patterns of the $FAPbI_3$ perovskite films without (C) and with (D) METEAM.

(E and F) The evolution of the (100) plane azimuth angle during annealing of perovskite films without (E) and with (F) METEAM.

(G) Azimuthal integrated curves of the (100) plane of the annealed $FAPbI_3$ perovskite films without and with METEAM.

(H and I) GIXRD patterns of the $FAPbI_3$ perovskite films without (H) and with (I) METEAM.

(J) XRD patterns of the $FAPbI_3$ perovskite films without and with METEAM additive.

The possible mechanism of METEAM-induced (100)-oriented crystallization is depicted in Figure 1F. Before thermal annealing, as revealed by *in situ* grazing incident XRD (GIXRD) in Figures 2A and 2B, METEAM promotes phase transformation from the non-photovoltaic yellow δ-phase to black-phase α-$FAPbI_3$, while the control film shows negligible existence of α-$FAPbI_3$. As the annealing progresses, the energy distribution of the phase transition process becomes relatively uniform in the control film, resulting in the coexistence of multiple phases and orientations.[23] The randomly distributed α-phase $FAPbI_3$ crystal nuclei in the target film turn into well-arranged under the driving force of S-Sn interactions between $METEAM^+$ and

SnO_2, and consequently act as the template to promote the phase transition leading to the formation of highly (100)-oriented $FAPbI_3$ perovskite film (Figures 1F and S6).

Effect of METEAM on the nucleation and crystal growth kinetics of perovskite

To verify the formation mechanism of METEAM-induced (100)-orientation $FAPbI_3$ perovskite film, we monitored the nucleation and crystallization process of $FAPbI_3$ perovskite films through *in situ* GIXRD analysis. The wet perovskite films were immediately transferred to a plate and heated from room temperature (~30°C) to 120°C for thermal annealing (heating rate: 120°C per min). The GIXRD patterns were recorded every 5 s. Figures 2A and 2B present the evolution of radially integrated intensities extracted from the *in situ* GIXRD patterns along with q = 7 to q = 12 nm^{-1} with the annealing time. At the initial stage, the control perovskite film shows fully hexagonal δ-phase $FAPbI_3$ with the diffraction signal at around q = 8.4 nm^{-1} (Figures 2A and S7A),[38,39] while the coexistence of hexagonal δ-phase $FAPbI_3$ and a small amount of cubic α-phase $FAPbI_3$ is observed in the target perovskite film (Figures 2B and S7B), indicating lowered formation energy of α-phase $FAPbI_3$ after METEAM incorporation. A plausible interpretation is that the $–H_3N^+$ head within $METEAM^+$ can anchor the Pb-I octahedrons, promoting their assembly upon evaporation of the solvent.[18,20,23,40]

Furthermore, the false-color intensity map versus the wave vector q_z (x axis) and the frame numbers (y axis) were plotted in Figures 2C and 2D to quantitatively study the crystallization process of the control and target perovskite films. The intensity of the (100) facets of the α-phase $FAPbI_3$ perovskite extracted from GIXRD is used to divide the different stages.[23,41] For the control perovskite film, nucleation and δ-phase formation in the initial stage during annealing are mainly divided into the nucleation stage (N_S) in which the disappearance, formation, and growth of the nuclei take place concurrently, followed by δ- to α-phase transformation occurred in the annealing process, as reflected by the rapid increase of the peak intensity of (100) facets. This stage is named as the rapid crystal growth stage G_I. In the last stage, the δ-phase is completely transformed into the α-phase, and the peak intensity of the α-phase tends to increase gradually, which is defined as the slow crystal growth stage G_{II} (Figure 2C). For the target perovskite film with METEAM ligand, the addition of METEAM ligand reduces the formation energy of the α-phase, thus there is already an α-phase in the unannealed state (i.e., wet film), as reflected by the intensity increase of (100) facets, and the δ- to α-phase transformation occurs already in the initial stage, followed by continuously rapid increase of the α-phase intensity. Hence, we define this rapid crystal growth stage as G_I. In the next stage, the δ- to α-phase transformation was finished completely like the control film, and the α-phase crystallization speed increased more slowly; thus, this slow crystal growth stage was labeled as G_{II} (Figure 2D).

Notably, the control film exhibits a faster increase in the (100) peak intensity during the G_I process than the target film with METEAM, indicating that METEAM leads to delayed crystal growth kinetics. This slower crystal growth may be attributed to the competition between $METEAM^+$ and FA^+ to interact with the Pb-I framework.[18] Besides, the *in situ* azimuth angle mapping of the (100) facets during the annealing stage (Figures 2E and 2F) was also extracted from the GIXRD patterns. The azimuth angle of the control film shows an approximately equal distribution around 45°, 90°, and 135° throughout the annealing process, indicating that the control film has a random crystallization orientation. However, a sharp peak centered at the azimuth angle of 90° is observed in the target film after about 60 s. These findings verify that METEAM incorporation deflects the (100) plane parallel to the substrate during the annealing stage via S-Sn interactions.

To further investigate the annealed control and target perovskite films, GIXRD characterizations of samples after annealing at 120°C for 1 h were performed. As shown in Figures 2H and 2I, the diffraction signals centered at around q = 10 and q = 20 nm^{-1} are assigned, respectively, to the (100) and (200) facets of α-$FAPbI_3$ perovskite (Figure S8).[6] The control perovskite film exhibits a Debye-Scherrer ring for (100) facets, indicating randomly distributed crystals. By contrast, discrete diffraction spots are observed in the target perovskite film, suggesting highly oriented crystals.[39,42] To scrutinize the final crystal plane orientation, we extracted the azimuthal scan profiles for the (100) facets from the GIXRD patterns of the annealed perovskite film (Figure 2G). The degree of such an in-plane orientation is then quantitatively evaluated via calculating the ratio of orientated crystallites according to the formula of $f_{ori} = \frac{A_{ori}}{A_{ori}+A_{iso}}$,[43] where A_{ori} and A_{iso} are the amount of oriented and isotropic crystallites, respectively, obtained from the integrated area of the corresponding fitted Gaussian peaks in Figure 2G. The f_{ori} of $FAPbI_3$ perovskite film increases significantly from 57.2% to 91.7% after METEAM incorporation, indicating that METEAM enables the formation of highly ordered $FAPbI_3$ perovskite film with the (100) plane parallel to the substrate.[44]

Top-surface scanning electron microscope (SEM) images and statistic distribution of grain sizes illustrated in Figure S9 indicate that the average grain size increases from 1.73 to 1.77 and 2.03 μm after 0.3 and 0.6 mol % METEAM incorporation, but decreases to 1.99 μm when excess METEAM (0.9 mol %) was incorporated. The increase in grain size is ascribed to the ligand-assisted perovskite crystallization process, and excess ligand affects perovskite crystallinity. Figures 2J and S10A compare the XRD patterns of the control and target perovskite films, showing two intense diffraction peaks located at 13.96° and 28.10° assigned to the (100) and (200) crystal planes of α-$FAPbI_3$ perovskite, respectively.[45] No shift of the diffraction peaks is observed, suggesting that METEAM does not enter into the $FAPbI_3$ crystal lattice (Figure S10B). Full width at half maxima (FWHM) was then extracted from XRD patterns to evaluate the crystallinity of perovskite. The obvious increment of the (100) crystal plane peak intensity for the target perovskite film is accompanied by the smallest FWHM obtained for the perovskite incorporated with 0.6 mol % METEAM (Figure S10C).[46,47] According to these results, we conclude that the optimal METEAM incorporation concentration is 0.6 mol %, under which high-quality perovskite film with the largest grain size and highest crystallinity is obtained.

Defect passivation of the bulk and buried interface of perovskite

To confirm the synergistic effects of the preferred facet orientation and the S-Pb coordination interactions on defect passivation, we first evaluated the trap state density (n_t) by using the space charge limited current (SCLC) method. The calculated n_t value within the $FAPbI_3$ perovskite film according to the dark current-voltage (*I-V*) curves of the electron-only devices (Figure 3A) decreases from 8.8 × 10^{14} (control) to 6.7 × 10^{14} cm^{-3} (target).[18,30] Furthermore, the lower trap state density translates into higher steady-state photoluminescence (PL) intensity and longer time-resolved PL lifetime (control, τ_{ave} = 1,105 ns; target, τ_{ave} = 1,426 ns) for the (100)-oriented $FAPbI_3$ perovskite film (Figures 3B and S11; Table S3), indicating the suppressed trap-assisted non-radiative carrier recombination.[48] In addition, the PL quantum yields (PLQYs) of the control and target perovskite films were also measured (Figure 3C). The PLQYs of the target film on the glass substrate exhibit a higher value than that of the control film, confirming the non-radiative recombination was suppressed. Besides, we also compare the PLQYs values of the control and target perovskite films deposited

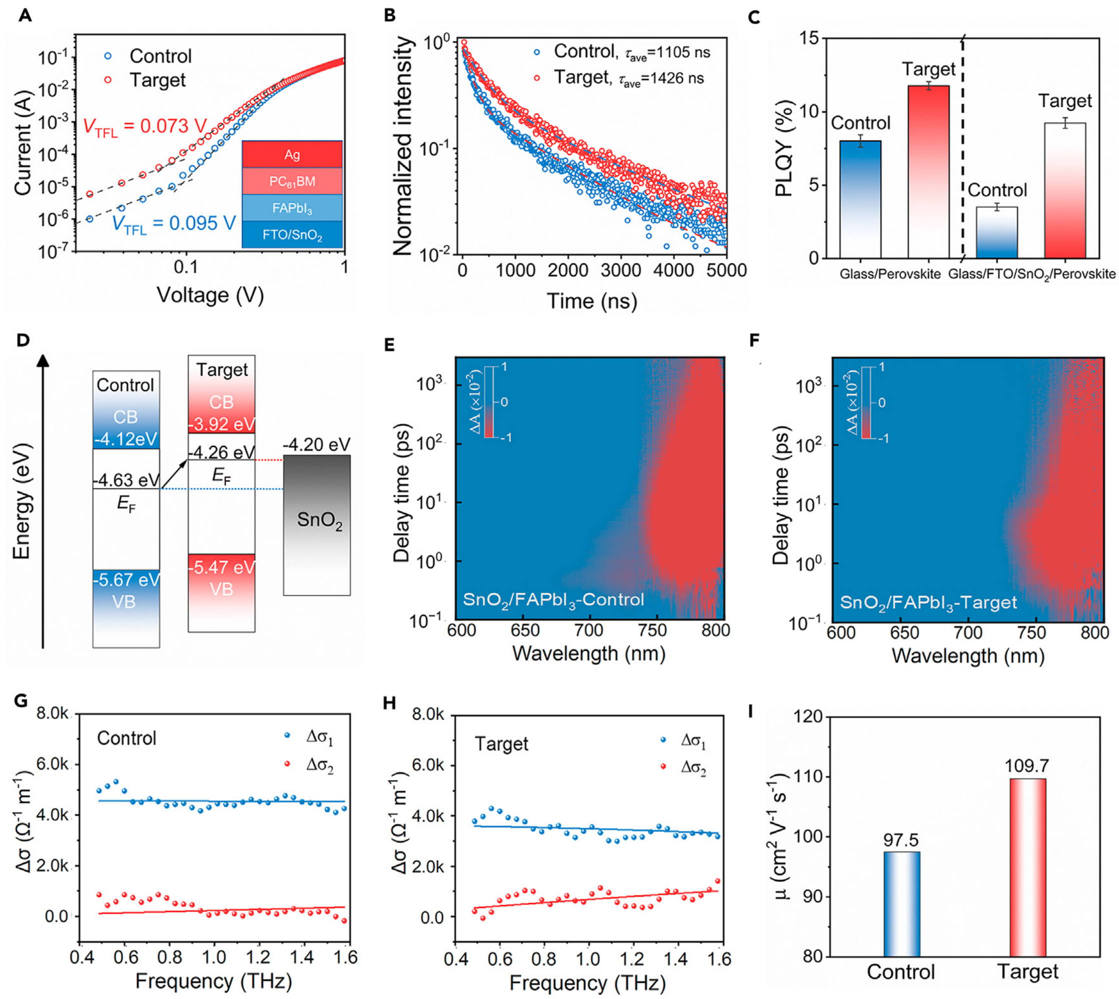

Figure 3. Passivation at the perovskite/ETL interface

(A) Dark current-voltage curves of the electron-only devices ($FTO/SnO_2/FAPbI_3/PC_{61}BM/Ag$) based on the $FAPbI_3$ perovskite films without and with METEAM, from which the trap-filled limit voltage (V_{TFL}) is determined as the kink point.

(B) Time-resolved PL of the $FAPbI_3$ perovskite films with and without METEAM deposited on the bare glass.

(C) PLQY of the perovskite film deposited onto the glass and glass/FTO/SnO_2 substrates.

(D) Energy-level diagram for the $FAPbI_3$ perovskite films without and with METEAM based on the parameters derived from UPS spectra.

(E and F) The pseudocolor plots of the glass/FTO/SnO_2/$FAPbI_3$ without (control) and with (target) METEAM excited at 400 nm and normalized decay kinetic curves at 778 nm of the TAS.

(G–I) Carrier mobilities (μ) extracted from the femtosecond-resolved OPTP spectroscopy of the $FAPbI_3$ perovskite films with and without METEAM deposited on Z-cut quartz substrate.

on glass/FTO/SnO_2 substrate and find that the PLQYs of the control film decreases from 8.0% to 3.5%, indicating the considerable non-radiative recombination at the control SnO_2/perovskite interface. By contrast, the gap of PLQYs values between the target film prepared on the glass and glass/FTO/SnO_2 substrates is much smaller (from 11.8% to 9.2%), implying suppressed non-radiative recombination benefited from the improved buried interface as discussed above.[49]

To unveil the influence of METEAM ligand on the microstructures of the buried interface, we first measured the cross-section SEM images of perovskite films. The target perovskite film shows larger grain sizes and reduced grain boundaries, and no discernible holes were found at the buried interface of both control and target perovskite films (Figure S12). Then, we carried out the SEM and atomic force microscopy (AFM) measurements on the buried interface from the perovskite side. The perovskite film gradually delaminates from the ETL-coated glass by exerting a constant force from one edge of the film and is peeled off together with epoxy and cover glass from the SnO_2/FTO substrate (Figure S13).[50] As shown in Figure S14, the control perovskite shows a relatively smooth buried interface covered with some small nanoparticle, which stemmed from the damaged integrity of the material, was compromised during the peeling off process. The stronger the adhesion between perovskite film and substrate, the more nanoparticles on the uncovered perovskite film. For the buried interface of the target perovskite film, the surface becomes rougher, and more nanoparticles are observed in the target perovskite film, indicating stronger interfacial contact between perovskite and SnO_2 after incorporating METEAM ligand.[12,25,50–52]

To investigate the effect of preferred orientation on the optoelectronic properties of $FAPbI_3$ perovskite, ultraviolet photoelectron spectroscopy (UPS) was implemented to probe the change of band structure of perovskite before and after METEAM incorporation (Figure S15). As shown in Figure 3D, the work function (WF) of the $FAPbI_3$ perovskite extracted from the secondary electron cutoff spectrum decreases from 4.63 to 4.26 eV after METEAM incorporation. Combined with the optical absorption measurements (Figure S16), the energy level diagram is depicted in Figure 3D. The Fermi level (E_F) of the target $FAPbI_3$ perovskite shifts closer to the conduction band minimum (E_C), suggesting that the (100)-oriented target film is more n-type characteristic despite that the band gap is unchanged (Table S4). This agrees well with the previous report and is expected to promote the charge transfer from the perovskite to SnO_2 ETL.[16–18] In addition, the shallower valence band maximum (E_V) leads to a larger energy level difference relative to the E_V of SnO_2 ETL, and this can prevent hole quenching at the ETL/perovskite interface.[45]

Ultrafast transient absorption (TA) characterization was next employed to gain a deeper insight into charge recombination dynamics. The pseudocolor plots of the glass/FTO/SnO_2/perovskite films excited at 400 nm (Figures 3E and 3F) along with the decay kinetic curves at 778 nm extracted from the ultrafast TA spectroscopy (TAS) demonstrate that the decay time ($\tau_{injection}$) for photoexcited electron injection from the $FAPbI_3$ perovskite to the SnO_2 ETL becomes shorter (from 125.5 ps for the control to 94.0 ps for the target) after METEAM incorporation (Figure S17; Table S5), indicating facilitated charge transfer at perovskite/SnO_2 interface.[44,53,54]

We further carried out the femtosecond-resolved optical-pump terahertz-probe (OPTP) spectroscopy measurements to obtain the carrier mobility of the $FAPbI_3$ perovskite without and with METEAM. The effective charge-carrier mobility increases from 97.5 to 109.7 $cm^2 V^{-1} s^{-1}$ for the $FAPbI_3$ perovskite after METEAM incorporation (Figures 3G–3I), and this is beneficial for the charge transfer within the perovskite layer.[48] Overall, based on a series of characterizations discussed above, we conclude that the preferred orientation of $FAPbI_3$ perovskite film and improved buried interface induced by METEAM result in enhanced optoelectronic properties of $FAPbI_3$ perovskite and can thus be used to realize high-performance PSCs.

Photovoltaic performance and stability

To evaluate the effect of METEAM incorporation on device performance, conventional-structure (n-i-p) devices with a configuration of FTO/SnO_2/perovskite/passivation

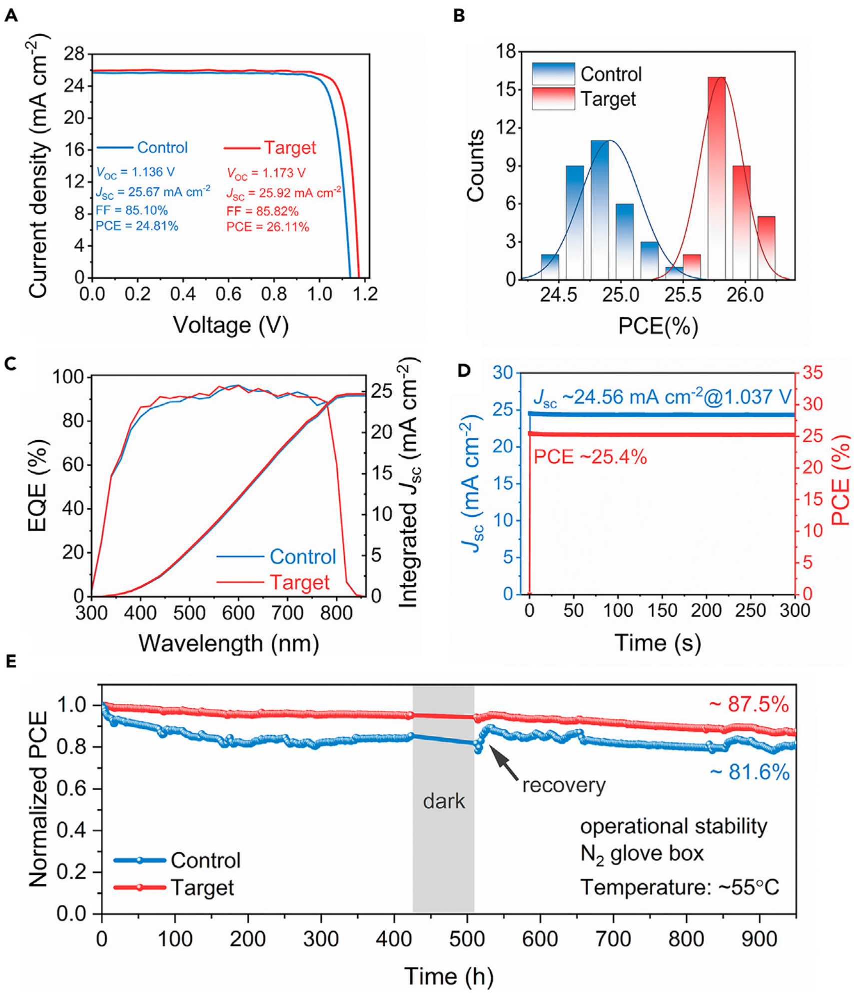

Figure 4. Device performance and stability

(A) *J-V* curves of the best-performing control and target devices.

(B) PCE distributions of the control and target PSC devices.

(C) EQE spectra of the best-performing control and target devices.

(D) The stabilized photocurrent and PCE output of the target device measured at the MPP.

(E) The evolution of normalized PCE of the unencapsulated device with the time under the LED illumination (100 mW cm^{-2}) in the N_2 glove box.

layer/2,2′,7,7′-Tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9′-spirobifluorene (spiro-OMeTAD)/Au were fabricated. Figure 4A presents the current density-voltage (*J-V*) curves of the best-performing control and target devices measured under standard air mass 1.5 global (AM 1.5 G) conditions. The control device delivers a PCE of 24.81%, with a V_{OC} of 1.136 V, a J_{SC} of 25.67 mA cm^{-2}, and a fill factor (FF) of 85.10%. Upon METEAM incorporation with the optimal concentration of 0.6 mol % (Table S6), the J_{SC}, V_{OC}, and FF values of the target device are 25.92 mA cm^{-2}, 1.173 V, and 85.82%, respectively, corresponding to a PCE of 26.11% with negligible hysteresis (Figure S18; Table S7). This value is among the highest PCE of conventional PSCs reported in the literature (Figure S19; Table S8). The target device was further verified by an independent solar cell-accredited laboratory (SIMIT, Shanghai, China), which realized a V_{OC} of 1.178 V, J_{SC} of 25.64 mA cm^{-2}, and an FF of 85.45%, contributing to a certified PCE of 25.80% (Figures S20 and S21; Table S9). The reproducibility was verified according to the photovoltaic performance parameters statistics based on 32 devices fabricated in the same batch (Figures 4B and S22), confirming the effectiveness of METEAM in enhancing device efficiency. The external quantum efficiency (EQE) spectra of the control and target devices are compared in Figure 4C, the slight enhancement of EQE value could be attributed to the passivation of the buried interface via S-Sn coordination.[55] Besides, the V_{OC} of the solar cell is determined by the electron and hole quasi-Fermi level splitting (QFLS) within the light-absorbing layer under non-equilibrium conditions.[56] The calculated QFLS of half-devices (glass/FTO/SnO_2/perovskite) without and with METEAM according to the corresponding PLQYs are 1.174 and 1.198 eV (Figure S23; Table S10), consistent with the increase of V_{OC}. In addition, the stabilized photocurrent and output of the target device at the maximum power point (MPP, V_{mpp}: 1.037 V) were measured for 300 s, and the target device exhibits a stabilized photocurrent of 24.56 mA cm^{-2} and PCE of 25.40% (Figure 4D).

The operational stability of the unencapsulated device under light-emitting diode (LED) illumination (100 mW cm^{-2}) for 950 h in a nitrogen-filled glovebox was also tracked. Given that the temperature of the device approaches near 55°C as measured by an infrared thermometer under illumination, spiro-OMeTAD is substituted with the more thermally stable poly[bis(4-phenyl) (2,4,6-trimethylphenyl) amine] (PTAA) for stability test.[57,58] As shown in Figure 4E, the control device suffers from a severe drop in the initial stage and recovers from the initial degradation when stored in the dark for around 45 h, whereas this phenomenon becomes not obvious in the target device due to the suppressed interfacial charge accumulation originated from the enhanced charge transfer as discussed above.[59] As a result, the target device maintains 87.5% of its initial PCE value, whereas only 81.6% of the original PCE is retained for the control device after 950 h illumination. In addition, the ambient stability tests were carried out at room temperature with a relative humidity of 35%. The unencapsulated target devices showed enhanced ambient stability than the control devices, retaining 93.8% of its initial PCE after nearly 1,000 h storage (Figure S24). Hence, in addition to PCE enhancement, METEAM incorporation leads to improved operational and ambient stabilities of PSC devices.

Conclusions

In summary, we develop a facile and effective strategy to concomitantly induce a well-contact buried interface and the (100)-oriented perovskite by incorporating a bifunctional ligand METEAM into perovskite precursor solution. METEAM forms a gradient vertical distribution across perovskite film, spontaneously aggregating at the buried interface, operating as a bifunctional bridge between the perovskite and SnO_2 ETL, and bidirectionally passivating their defects. Moreover, METEAM molecules preferentially adsorb on (100) facets of perovskite via their strong interactions with perovskite lattice, thereby decreasing the surface energy and inducing oriented perovskite crystallization. After METEAM incorporation, $FAPbI_3$ perovskite films exhibit lower trap state density, longer carrier diffusion lifetime, and higher carrier mobility, which are beneficial for interfacial charge transfer with suppressed non-radiative recombination. The resultant conventional-structure PSC devices deliver a PCE of 26.1% (certified 25.8%), which is among the highest PCE of conventional PSCs, and the operational stability of PSC devices is also improved obviously. Concomitant regulation of crystallization orientation and buried interface via ligand engineering fulfilled in this work offers an applicable route for efficiency enhancement of PSCs toward commercialization.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Shangfeng Yang (sfyang@ustc.edu.cn).

Materials availability

This study did not generate new unique materials.

Data and code availability

This study did not generate or analyze any datasets and codes.

All data are available in the main text or the supplemental information.

Materials

All chemicals were used as received without further purification, including urea (Sigma-Aldrich, ≥99%), hydrochloric acid (HCl, 37 wt % in water, Sinopharm Chemical Reagent), thioglycolic acid (Sigma-Aldrich, 98%), $SnCl_2 \cdot 2H_2O$ (Aladdin, ≥99.99%), KCl (Aladdin, ≥99.99%), PbI_2 (TCI, 99.99%), methylammonium chlorine (MACl, Xi'an Polymer Light Technology Corporation), FAI (Dyesol), rubidium chloride (Adamas, >99.5%), METEAM (Alfa Aesar, ≥97%), hexylamine hydrobromide (TCI, >98%), spiro-OMeTAD (1M company), 4-tert-butylpyridine (t-BP, Adamas, >97%), li-bis(trifluoromethanesulfonyl)imide (Li-TFSI, Sigma-Aldrich), pyrazol-1-yl)-4-tert-butylpyridine)-cobalt(III)Tris(bis(trifluoromethylsulfonyl)imide)) salt (Co(III) TFSI, Dyesol), PTAA (Xi'an Polymer Light Technology Corporation), and 4-isopropyl-4′-methyldiphenyliodonium Tetrakis(pentafluorophenyl)borate (TPFB, TCI, 98%). Dimethylformamide (DMF), dimethyl sulfoxide (DMSO), chlorobenzene (CB), and acetonitrile (ACN) were all purchased from Sigma-Aldrich. Isopropyl alcohol (IPA), chloroform, and diethyl ether were all purchased from Sinopharm Chemical Reagent.

Device fabrication

The FTO glass substrates (1.5 × 1.5 cm^2) were cleaned by sequential ultrasonic treatment in detergent, deionized water, acetone, and IPA for 15 min each. Then, the substrates were dried by nitrogen flow. Before fabricating the SnO_2 layer, the substrates were treated by UV-ozone for 30 min. The SnO_2 layer was fabricated by CBD method according to the previous reports.[44,45] Briefly, the solution was prepared by dissolving 1,250 mg of urea in 100 mL deionized water, followed by adding 1,250 μL HCl and 25 μL of thioglycolic acid. Finally, 275 mg $SnCl_2 \cdot 2H_2O$ was added to the solution followed by stirring for 2 min. Then, the FTO substrates, which were taped with a kapton tape on the edge to prevent deposition of SnO_2, were vertically placed inside the glassware that contained about 100 mL of the solution. The glassware was placed inside a water bath set to 94°C. After 4 h reaction, the FTO/SnO_2 substrates were removed from the glassware and cleaned via sonication with deionized water and IPA for 5 min each and then dried by nitrogen flow. The FTO/SnO_2 substrates were annealed in an ambient environment (RH ∼ 25%) at 170°C for 1 h. After cooling to room temperature, the substrates were treated by UV-ozone for 15 min, followed by spin coating 10 mM KCl in deionized water at 3,000 rpm for 30 s and annealing at 100°C for 10 min. Before the perovskite deposition, the KCl-modified FTO/SnO_2 substrates were treated by UV-ozone for 15 min and then pumped inside a nitrogen glovebox.

The perovskite precursor solution was prepared by mixing 705.34 mg PbI_2, 240.76 mg FAI, 33.76 mg MACl, and 4 mg RbCl in 1 mL mixed solvents

(DMF:DMSO = 8:1, v:v). The perovskite precursor solution containing METEAM was prepared by adding 0.6 mol % METEAM into the solution. Then, the perovskite precursor solution was deposited via spin-coating at 1,000 rpm for 10 s and 5,000 rpm for 30 s, both with the ramp of 2,000 rpm/s. 700 μL of diethyl ether solution was deposited onto the substrate as 20 s left of the procedure. Subsequently, the FTO/SnO_2/perovskite substrate was annealed at 120°C for 1 h. After the substrate was cooled to room temperature, 10 mM of n-hexylammonium bromide chloroform solution was deposited onto the perovskite layer at 3,000 rpm for 30 s and annealed at 100°C for 5 min.[60] The hole transporting layer (HTL) solution, which contains 100 mg spiro-OMeTAD, 39 μL t-BP, 10 μL Co (III) TFSI solution (0.25 M in ACN), 23 μL Li-TFSI solution (1.8 M in ACN), and 1,095 μL CB, was spin-coated onto the n-hexylammonium bromide-treated perovskite film at 3,000 rpm for 30 s with the ramp of 3,000 rpm/s. For the stability test, PTAA doped with TPFB was used to replace spiro-OMeTAD as the HTL considering that the spiro-OMeTAD was not a stable HTL. The concentration of PTAA was 30 mg mL^{-1} in toluene, and the weight ratio of PTAA/TPFB was 10:1. The PTAA was deposited on top of the n-hexylammonium bromide-treated perovskite layer at a spin rate of 2,000 rpm for 30 s. Finally, 65 nm gold was deposited via a shadow mask at a pressure of about 2×10^{-4} Pa in a vacuum chamber. The deposition rate of Au maintained 0.1 Å/s for the first 5 nm and then changed to 0.2 Å/s to deposit 5–10 nm, finally the rate changed to 0.4 Å/s and maintained until the end of deposition process. The device size areas were 0.1369 cm^2 (0.37×0.37 cm^2). When measuring, a 0.0905 cm^2 (0.3×0.3 cm^2) non-reflective metal mask aperture was used to define the accurate active area of the devices.

Characterization

The J-V characterizations were measured by a Keithley 2400 source measurement unit under simulated AM 1.5 irradiation (100 mW cm^{-2}) with a standard xenon-lamp-based solar simulator (Oriel Sol 3A, USA), which was calibrated with a monocrystalline silicon reference cell (Oriel P/N 91150 V, with KG-5 visible color filter) calibrated by the National Renewable Energy Laboratory (NREL). The active device area during the measurements was 0.0905 cm^2 as defined by a metal mask. The EQE measurements were carried out on an ORIEL Intelligent Quantum Efficiency (IQE) 200TM Measurement system established with a tunable light source. SEM measurement was attained via a field-emission SEM (FEI Apero). AFM images were carried out on a Dimension EDGE scanning probe microscope in Peak Force Tapping mode (Bruker). The XRD patterns were obtained on a Rigaku SmartLab X-ray diffractometer with Cu-Kα radiation (0.154 nm). UV-vis spectroscopy was recorded on a UV-vis-NIR 3600 spectrometer (Shimadzu, Japan). XPS measurements were performed on a Thermo ESCALAB 250 instrument with a monochromatized Al Kα X-ray source in vacuum. The UPS experiments were performed at the beamline BL11U in the National Synchrotron Radiation Laboratory (NSRL), Hefei. The steady-state PL spectra were recorded by employing an Edinburgh Instruments FLS920 fluorescence spectrometer with an excitation wavelength of 500 nm. PLQY data were measured using a Horiba Fluorolog-3 system with a petite integrating sphere. The excitation wavelength was 460 nm, which was provided by a semiconductor laser (DL-405-100-T, Raymount Optronics). Time-resolved PL spectra were measured via the time-correlated single-photon counting method with a PicoQuant Gmbh Solea Supercontinuum Laser. A picosecond pulsed diode laser at 543 nm with a pulse width of 104 ps was used as the excitation source. The ultrafast TAS was performed on a pump-probe system (Helios, Ultrafast System) with the maximum time delay of ~8 ns using a motorized optical delay line under ambient conditions. The pump pulses at 400 nm (~20 μW average power at the sample) were delivered by an ultrafast optical parametric amplifier (Opera Solo) excited by a regenerative amplifier (Coherent Astrella, 800 nm, 35 fs, 5 mJ, 1 kHz), seeded with a mode-locked Ti:sapphire oscillator (Coherent Vitara, 800 nm, 80 MHz), and pumped with a LBO laser (Coherent Evolution-50C, 1 kHz system). In the time-resolved OPTP spectroscopy experiment, a Ti:sapphire laser with a pulse width of 35 fs, a center wavelength of 800 nm, and a repetition rate of 1 kHz was used to generate the pump pulse. The output laser energy was divided into three beams, used for terahertz (THz) generation, THz detection, and optical pump of our samples, respectively. THz pulses for probe were generated by 0.5 mm-thick (110)-oriented ZnTe crystal and were focused into THz beam by off-axis-parabolic (OAP) mirrors then through the sample. The THz beam through the sample was collected by a pair of OAP mirrors and detected by electro-optic sampling. The optical-pump pulse is focused on the sample with a diameter of 6 mm. All the measurements were performed under 5% humidity by purging the THz beam path with dry air to prevent water vapor absorption in the ambient air. ToF-SIMS 5 system (ION-TOF) was used to obtain the ToF-SIMS depth profiling data. Bi^{3+} ions were used as primary ions and negative ions were detected. Sputtering was performed using Cs^{+} sputtering ions with 500 eV ion energy, 50 nA ion current, and a 200 × 200 μm^2 raster size. An area of 50 × 50 μm^2 was analyzed using Bi^{3+} ions with 30 keV ion energy.

DFT simulations

The first-principle DFT simulations were performed with the Vienna Ab Initio Simulation Package (vasp 6.4) to study the geometric and electronic structures of all the surface perovskite derivative structures with METEAM. The projector augmented wave (PAW) with the cutoff energy of 600 eV was employed. The generalized gradient approximation (GGA) exchange-correlation functional of Perdew-Burke-Ernzerhof (PBE) was adopted in the DFT calculations.[61] The electronic constituents are 5d 6s 6p for Pb, 5p 6s for I, 3s 3p for S, 2s 2p for C and N, and 1s for H. For the surface structures, we adopted a 2 × 2 × 1 Γ-centered k-point grid, generated by the Monkhorst-Pack scheme, for detailed properties obtained with PBE functional. Considering the interaction between the hydrogen atoms and high-electronegativity groups, the PBE with the DFT-D3 dispersion correction of Grimme with zero-damping was applied to optimize the geometric structures.[62–64] During the optimization of the geometries, all structures were allowed to relax to ensure that each atom was in mechanical equilibrium without any residual force larger than 10^{-4} eV/Å, except the bottom layers were fixed.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2024.07.009.

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AUTHOR CONTRIBUTIONS

S.Y. supervised the overall project. Xingcheng Li designed and conducted device fabrication. S.G. and Xinyu Li assisted in the device fabrication. Q.L., L.Z., Z. Li, and X.Z. helped to conduct the DFT calculations. Y.W. assisted in OPTP measurement. Z. Liu, Q.H., and Y.L. helped to analyze the OPTP data. C.W. and X.G. assisted in the GIXRD measurement. P.X. and T.C. helped to conduct the TAS characterization. W.C. and Z.X. helped to carry out the PLQY measurement. Xingcheng Li and S.G. drafted the manuscript under the guidance of S.Y. All the authors revised the manuscript. S.Y., X.W., Z.Z., and S.X. finalized the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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REFERENCES

1. Dong, Q.F., Fang, Y.J., Shao, Y.C., Mulligan, P., Qiu, J., Cao, L., and Huang, J.S. (2015). Electron-hole diffusion lengths >175 μm in solution-grown $CH_3NH_3PbI_3$ single crystals. Science *347*, 967–970. https://doi.org/10.1126/science.aaa5760.

2. Xing, G.C., Mathews, N., Sun, S.Y., Lim, S.S., Lam, Y.M., Grätzel, M., Mhaisalkar, S., and Sum, T.C. (2013). Long-range balanced electron- and hole-transport lengths in organic-inorganic $CH_3NH_3PbI_3$. Science *342*, 344–347. https://doi.org/10.1126/science.1243167.

3. Eperon, G.E., Stranks, S.D., Menelaou, C., Johnston, M.B., Herz, L.M., and Snaith, H.J. (2014). Formamidinium lead trihalide: a broadly tunable perovskite for efficient planar heterojunction solar cells. Energy Environ. Sci. *7*, 982–988. https://doi.org/10.1039/c3ee43822h.

4. De Wolf, S., Holovsky, J., Moon, S.-J., Löper, P., Niesen, B., Ledinsky, M., Haug, F.-J., Yum, J.-H., and Ballif, C. (2014). Organometallic halide perovskites: sharp optical absorption edge and its relation to photovoltaic performance. J. Phys. Chem. Lett. *5*, 1035–1039. https://doi.org/10.1021/jz500279b.

5. NREL (2024). Best research-cell efficiency chart. https://www.nrel.gov/pv/cell-efficiency.html.

6. Jeong, J., Kim, M., Seo, J., Lu, H.Z., Ahlawat, P., Mishra, A., Yang, Y.G., Hope, M.A., Eickemeyer, F.T., Kim, M., et al. (2021). Pseudo-halide anion engineering for alpha-$FAPbI_3$ perovskite solar cells. Nature *592*, 381–385. https://doi.org/10.1038/s41586-021-03406-5.

7. Min, H., Lee, D.Y., Kim, J., Kim, G., Lee, K.S., Kim, J., Paik, M.J., Kim, Y.K., Kim, K.S., Kim, M.G., et al. (2021). Perovskite solar cells with atomically coherent interlayers on SnO_2 electrodes. Nature *598*, 444–450. https://doi.org/10.1038/s41586-021-03964-8.

8. Kim, M., Jeong, J., Lu, H.Z., Lee, T.K., Eickemeyer, F.T., Liu, Y.H., Choi, I.W., Choi, S.J., Jo, Y., Kim, H.B., et al. (2022). Conformal quantum dot-SnO_2 layers as electron transporters for efficient perovskite solar cells. Science *375*, 302–306. https://doi.org/10.1126/science.abh1885.

9. Zhao, Y., Ma, F., Qu, Z.H., Yu, S.Q., Shen, T., Deng, H.X., Chu, X.B., Peng, X.X., Yuan, Y.B., Zhang, X.W., et al. (2022). Inactive $(PbI_2)_2RbCl$ stabilizes perovskite films for efficient solar cells. Science *377*, 531–534. https://doi.org/10.1126/science.abp8873.

10. Park, J., Kim, J., Yun, H.-S., Paik, M.J., Noh, E., Mun, H.J., Kim, M.G., Shin, T.J., and Seok, S.I. (2023). Controlled growth of perovskite layers with volatile alkylammonium chlorides. Nature *616*, 724–730. https://doi.org/10.1038/s41586-023-05825-y.

11. Peng, W., Mao, K., Cai, F., Meng, H., Zhu, Z., Li, T., Yuan, S., Xu, Z., Feng, X., Xu, J., et al. (2023). Reducing nonradiative recombination in perovskite solar cells with a porous insulator contact. Science *379*, 683–690. https://doi.org/10.1126/science.ade3126.

12. Tan, Q., Li, Z., Luo, G., Zhang, X., Che, B., Chen, G., Gao, H., He, D., Ma, G., Wang, J., et al. (2023). Inverted perovskite solar cells using dimethylacridine-based dopants. Nature *620*, 545–551. https://doi.org/10.1038/s41586-023-06207-0.

13. Kojima, A., Teshima, K., Shirai, Y., and Miyasaka, T. (2009). Organometal halide perovskites as visible-light sensitizers for photovoltaic cells. J. Am. Chem. Soc. *131*, 6050–6051. https://doi.org/10.1021/ja809598r.

14. Guillemoles, J.-F., Kirchartz, T., Cahen, D., and Rau, U. (2019). Guide for the perplexed to the Shockley–Queisser model for solar cells. Nat. Photonics *13*, 501–505. https://doi.org/10.1038/s41566-019-0479-2.

15. Ma, C., Grätzel, M., and Park, N.-G. (2022). Facet engineering for stable, efficient perovskite solar cells. ACS Energy Lett. *7*, 3120–3128. https://doi.org/10.1021/acsenergylett.2c01623.

16. Ma, C., Kang, M.-C., Lee, S.-H., Kwon, S.J., Cha, H.-W., Yang, C.-W., and Park, N.-G. (2022). Photovoltaically top-performing perovskite crystal facets. Joule *6*, 2626–2643. https://doi.org/10.1016/j.joule.2022.09.012.

17. Lin, Q., Armin, A., Nagiri, R.C.R., Burn, P.L., and Meredith, P. (2015). Electro-optics of perovskite solar cells. Nat. Photonics *9*, 106–112. https://doi.org/10.1038/nphoton.2014.284.

18. Zheng, X.P., Hou, Y., Bao, C.X., Yin, J., Yuan, F.L., Huang, Z.R., Song, K.P., Liu, J.K., Troughton, J., Gasparini, N., et al. (2020). Managing grains and interfaces via ligand anchoring enables 22.3%-efficiency inverted perovskite solar cells. Nat. Energy *5*, 131–140. https://doi.org/10.1038/s41560-019-0538-4.

19. Ma, C., Eickemeyer, F.T., Lee, S.-H., Kang, D.-H., Kwon, S.J., Grätzel, M., and Park, N.-G. (2023). Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells. Science *379*, 173–178. https://doi.org/10.1126/science.adf3349.

20. Luo, C., Zheng, G.H., Gao, F., Wang, X.J., Zhao, Y., Gao, X.Y., and Zhao, Q. (2022). Facet orientation tailoring via 2D-seed-induced growth enables highly efficient and stable perovskite solar cells. Joule *6*, 240–257. https://doi.org/10.1016/j.joule.2021.12.006.

21. Li, M., Sun, R., Chang, J., Dong, J., Tian, Q., Wang, H., Li, Z., Yang, P., Shi, H., Yang, C., et al. (2023). Orientated crystallization of FA-based perovskite via hydrogen-bonded polymer network for efficient and stable solar cells. Nat. Commun. *14*, 573. https://doi.org/10.1038/s41467-023-36224-6.

22. Li, G., Su, Z., Canil, L., Hughes, D., Aldamasy, M.H., Dagar, J., Trofimov, S., Wang, L., Zuo, W., Jerónimo-Rendon, J.J., et al. (2023). Highly efficient p-i-n perovskite solar cells that endure temperature variations. Science *379*, 399–403. https://doi.org/10.1126/science.add7331.

23. Shi, P., Ding, Y., Ding, B., Xing, Q., Kodalle, T., Sutter-Fella, C.M., Yavuz, I., Yao, C., Fan, W., Xu, J., et al. (2023). Oriented nucleation in formamidinium perovskite for photovoltaics. Nature *620*, 323–327. https://doi.org/10.1038/s41586-023-06208-z.

24. Jiang, X., Liu, B., Wu, X., Zhang, S., Zhang, D., Wang, X., Gao, S., Huang, Z., Wang, H., Li, B., et al. (2024). Top-down Induced Crystallization Orientation toward Highly Efficient p-i-n perovskite Solar Cells. Adv. Mater. *36*, e2313524. https://doi.org/10.1002/adma.202313524.

25. Yang, X., Luo, D., Xiang, Y., Zhao, L., Anaya, M., Shen, Y., Wu, J., Yang, W., Chiang, Y.H., Tu, Y., et al. (2021). Buried interfaces in halide perovskite photovoltaics. Adv. Mater. *33*, e2006435. https://doi.org/10.1002/adma.202006435.

26. Ji, X., Bi, L., Fu, Q., Li, B., Wang, J., Jeong, S.Y., Feng, K., Ma, S., Liao, Q., Lin, F.R., et al. (2023). Target therapy for buried interface enables stable perovskite solar cells with 25.05% efficiency. Adv. Mater. *35*, e2303665. https://doi.org/10.1002/adma.202303665.

27. Luo, C., Zheng, G., Gao, F., Wang, X., Zhan, C., Gao, X., and Zhao, Q. (2023). Engineering the buried interface in perovskite solar cells via lattice-matched electron transport layer. Nat. Photonics *17*, 856–864. https://doi.org/10.1038/s41566-023-01247-4.

28. Lee, J.H., Lee, S., Kim, T., Ahn, H., Jang, G.Y., Kim, K.H., Cho, Y.J., Zhang, K., Park, J.-S., and Park, J.H. (2023). Interfacial α-$FAPbI_3$ phase stabilization by reducing oxygen vacancies in SnO_{2-x}. Joule *7*, 380–397. https://doi.org/10.1016/j.joule.2022.12.006.

29. Wang, Z., Kamarudin, M.A., Huey, N.C., Yang, F., Pandey, M., Kapil, G., Ma, T.L., and Hayase, S. (2018). Interfacial sulfur functionalization anchoring SnO_2 and $CH_3NH_3PbI_3$ for enhanced stability and trap passivation in perovskite solar cells. ChemSusChem *11*, 3941–3948. https://doi.org/10.1002/cssc.201801888.

30. Xia, M.H., Xia, Y.K., Zheng, Z.W., Chen, X.Y., Yang, J., Du, J.K., Mei, A.Y., Hu, Y., and Han, H.W. (2022). Defect passivation by a sulfur-containing Lewis base for efficient printable mesoscopic perovskite solar cells. Sol. Rrl *6*, 2200802. https://doi.org/10.1002/solr.202200802.

31. Yang, L., Zhou, H., Duan, Y., Wu, M., He, K., Li, Y., Xu, D., Zou, H., Yang, S., Fang, Z., et al. (2023). 25.24%-Efficiency FACsPbI3 Perovskite Solar Cells Enabled by Intermolecular Esterification Reaction of DL-Carnitine Hydrochloride. Adv. Mater. *35*, e2211545. https://doi.org/10.1002/adma.202211545.

32. Yang, D., Yang, R., Wang, K., Wu, C., Zhu, X., Feng, J., Ren, X., Fang, G., Priya, S., and Liu, S.F. (2018). High efficiency planar-type perovskite solar cells with negligible hysteresis using EDTA-complexed SnO_2. Nat. Commun. *9*, 3239. https://doi.org/10.1038/s41467-018-05760-x.

33. Guo, H.D., Xiang, W.H., Fang, Y.Y., Li, J.R., and Lin, Y. (2023). Molecular bridge on buried interface for efficient and stable perovskite solar cells. Angew. Chem. Int. Ed. Engl. *62*, e202304568. https://doi.org/10.1002/anie.202304568.

34. Zhang, Y., Wang, Y., Yang, X., Zhao, L., Su, R., Wu, J., Luo, D., Li, S., Chen, P., Yu, M., et al. (2021). Mechanochemistry advances high-performance perovskite solar cells. Adv. Mater. *34*, 2107420. https://doi.org/10.1002/adma.202107420.

35. Chen, A.Z., Shiu, M., Ma, J.H., Alpert, M.R., Zhang, D., Foley, B.J., Smilgies, D.-M., Lee, S.-H., and Choi, J.J. (2018). Origin of vertical orientation in two-dimensional metal halide perovskites and its effect on photovoltaic performance. Nat. Commun. *9*, 1336. https://doi.org/10.1038/s41467-018-03757-0.

36. Yan, K., Long, M., Zhang, T., Wei, Z., Chen, H., Yang, S., and Xu, J. (2015). Hybrid halide perovskite solar cell precursors: colloidal chemistry and coordination engineering behind device processing for high efficiency. J. Am. Chem. Soc. *137*, 4460–4468. https://doi.org/10.1021/jacs.5b00321.

37. Liu, J.L., Li, S., Liu, S., Chu, Y.M., Ye, T., Qiu, C., Qiu, Z.X., Wang, X.D., Wang, Y.F., Su, Y.Q., et al. (2022). Oxygen vacancy management for high-temperature mesoporous SnO_2 electron transport layers in printable perovskite solar cells. Angew. Chem. Int. Ed. Engl. *61*, e202202012. https://doi.org/10.1002/anie.202202012.

38. Gratia, P., Zimmermann, I., Schouwink, P., Yum, J.H., Audinot, J.N., Sivula, K., Wirtz, T., and Nazeeruddin, M.K. (2017). The many faces of mixed ion perovskites: unraveling and understanding the crystallization process. ACS Energy Lett. *2*, 2686–2693. https://doi.org/10.1021/acsenergylett.7b00981.

39. Park, B.W., Kwon, H.W., Lee, Y., Lee, D.Y., Kim, M.G., Kim, G., Kim, K.J., Kim, Y.K., Im, J., Shin, T.J., et al. (2021). Stabilization of formamidinium lead triiodide alpha-phase with isopropylammonium chloride for perovskite solar cells. Nat. Energy *6*, 419–428. https://doi.org/10.1038/s41560-021-00802-z.

40. Ren, H., Yu, S.D., Chao, L.F., Xia, Y.D., Sun, Y.H., Zuo, S.W., Li, F., Niu, T., Yang, Y., Ju, H., et al. (2020). Efficient and stable Ruddlesden-Popper perovskite solar cell with tailored interlayer molecular interaction. Nat. Photonics *14*, 154–163. https://doi.org/10.1038/s41566-019-0572-6.

41. Huang, Z., Bai, Y., Huang, X., Li, J., Wu, Y., Chen, Y., Li, K., Niu, X., Li, N., Liu, G., et al. (2023). Anion–π interactions suppress phase impurities in $FAPbI_3$ solar cells. Nature *623*, 531–537. https://doi.org/10.1038/s41586-023-06637-w.

42. Zhou, W.R., Li, D., Xiao, Z.G., Wen, Z.L., Zhang, M.M., Hu, W.P., Wu, X.J., Wang, M.T., Zhang, W.H., Lu, Y.L., et al. (2019). Zwitterion coordination induced highly orientational order of $CH_3NH_3PbI_3$ perovskite film delivers a high open circuit voltage exceeding 1.2 V. Adv. Funct. Mater. *29*, 1901026. https://doi.org/10.1002/adfm.201901026.

43. Zhang, Y.Z., Wang, Y.J., Yang, X.Y., Zhao, L.C., Su, R., Wu, J., Luo, D.Y., Li, S.D., Chen, P., Yu, M.T., et al. (2022). Mechanochemistry advances high-performance perovskite solar cells. Adv. Mater. *34*, e2107420. https://doi.org/10.1002/adma.202107420.

44. Li, X.C., Wu, X., Li, B., Cen, Z.Y., Shang, Y.B., Lian, W.T., Cao, R., Jia, L.B., Li, Z., Gao, D.P., et al. (2022). Modulating the deep-level defects and charge extraction for efficient perovskite solar cells with high fill factor over 86%. Energy Environ. Sci. *15*, 4813–4822. https://doi.org/10.1039/D2EE02543D.

45. Yoo, J.J., Seo, G., Chua, M.R., Park, T.G., Lu, Y.L., Rotermund, F., Kim, Y.K., Moon, C.S., Jeon, N.J., Correa-Baena, J.P., et al. (2021). Efficient perovskite solar cells via improved carrier management. Nature *590*, 587–593. https://doi.org/10.1038/s41586-021-03285-w.

46. Kim, G., Min, H., Lee, K.S., Lee, D.Y., Yoon, S.M., and Seok, S.I. (2020). Impact of strain relaxation on performance of alpha-formamidinium lead iodide perovskite solar cells. Science *370*, 108–112. https://doi.org/10.1126/science.abc4417.

47. Castro-Méndez, A.F., Hidalgo, J., and Correa-Baena, J.P. (2019). The role of grain boundaries in perovskite solar cells. Adv. Energy Mater. *9*, 1901489. https://doi.org/10.1002/aenm.201901489.

48. Lin, R.X., Xu, J., Wei, M.Y., Wang, Y.R., Qin, Z.Y., Liu, Z., Wu, J.L., Xiao, K., Chen, B., Park, S.M., et al. (2022). All-perovskite tandem solar cells with improved grain surface passivation. Nature *603*, 73–78. https://doi.org/10.1038/s41586-021-04372-8.

49. Chen, H., Maxwell, A., Li, C., Teale, S., Chen, B., Zhu, T., Ugur, E., Harrison, G., Grater, L., Wang, J., et al. (2023). Regulating surface potential maximizes voltage in all-perovskite tandems. Nature *613*, 676–681. https://doi.org/10.1038/s41586-022-05541-z.

50. Hao, M., Duan, T., Ma, Z., Ju, M.G., Bennett, J.A., Liu, T., Guo, P., and Zhou, Y. (2023). Flattening grain-boundary grooves for perovskite solar cells with high Optomechanical reliability. Adv. Mater. *35*, e2211155. https://doi.org/10.1002/adma.202211155.

51. Gao, H., Xiao, K., Lin, R.X., Zhao, S.Y., Wang, W.L., Dayneko, S., Duan, C.Y., Ji, C.L., Sun, H.F., Bui, A.D., et al. (2024). Homogeneous crystallization and buried interface passivation for perovskite tandem solar modules. Science *383*, 855–859. https://doi.org/10.1126/science.adj6088.

52. Zhou, Y., Herz, L.M., Jen, A.K.Y., and Saliba, M. (2022). Advances and challenges in

understanding the microscopic structure–property–performance relationship in perovskite solar cells. Nat. Energy *7*, 794–807. https://doi.org/10.1038/s41560-022-01096-5.

53. Jiang, X., Wang, X., Wu, X., Zhang, S., Liu, B., Zhang, D., Li, B., Xiao, P., Xu, F., Lu, H., et al. (2023). Strain regulation via pseudo halide-based ionic liquid toward efficient and stable α-$FAPbI_3$ inverted perovskite solar cells. Adv. Energy Mater *13*, 2300700. https://doi.org/10.1002/aenm.202300700.

54. Zhang, J., Wang, L., Jiang, C., Cheng, B., Chen, T., and Yu, J. (2021). $CsPbBr_3$ nanocrystal induced bilateral interface modification for efficient planar perovskite solar cells. Adv. Sci. (Weinh) *8*, e2102648. https://doi.org/10.1002/advs.202102648.

55. Quantum efficiency. https://www.pveducation.org/pvcdrom/solar-cell-operation/quantum-efficiency.

56. Chen, B., Rudd, P.N., Yang, S., Yuan, Y.B., and Huang, J.S. (2019). Imperfections and their passivation in halide perovskite solar cells. Chem. Soc. Rev. *48*, 3842–3867. https://doi.org/10.1039/c8cs00853a.

57. Luo, J.S., Xia, J.X., Yang, H., Chen, L.L., Wan, Z.Q., Han, F., Malik, H.A., Zhu, X.H., and Jia, C.Y. (2018). Toward high-efficiency, hysteresis-less, stable perovskite solar cells: unusual doping of a hole-transporting material using a fluorine-containing hydrophobic Lewis acid. Energy Environ. Sci. *11*, 2035–2045. https://doi.org/10.1039/C8EE00036K.

58. Jiang, Q., Zhao, Y., Zhang, X.W., Yang, X.L., Chen, Y., Chu, Z.M., Ye, Q.F., Li, X.X., Yin, Z.G., and You, J.B. (2019). Surface passivation of perovskite film for efficient solar cells. Nat. Photonics *13*, 460–466. https://doi.org/10.1038/s41566-019-0398-2.

59. Zhang, Z., Wang, H., Jacobsson, T.J., and Luo, J. (2022). Big data driven perovskite solar cell stability analysis. Nat. Commun. *13*, 7639. https://doi.org/10.1038/s41467-022-35400-4.

60. Yoo, J.J., Wieghold, S., Sponseller, M.C., Chua, M.R., Bertram, S.N., Hartono, N.T.P., Tresback, J.S., Hansen, E.C., Correa-Baena, J.-P., Bulović, V., et al. (2019). An interface stabilized perovskite solar cell with high stabilized efficiency and low voltage loss. Energy Environ. Sci. *12*, 2192–2199. https://doi.org/10.1039/C9EE00751B.

61. Perdew, J.P., Burke, K., and Ernzerhof, M. (1996). Generalized gradient approximation made simple. Phys. Rev. Lett. *77*, 3865–3868. https://doi.org/10.1103/PhysRevLett.77.3865.

62. Grimme, S. (2006). Semiempirical GGA-type density functional constructed with a long-range dispersion correction. J. Comp. Chem. *27*, 1787–1799. https://doi.org/10.1002/jcc.20495.

63. Grimme, S., Antony, J., Ehrlich, S., and Krieg, H. (2010). A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu. J. Chem. Phys. *132*, 154104. https://doi.org/10.1063/1.3382344.

64. Moellmann, J., and Grimme, S. (2014). DFT-D3 study of some molecular crystals. J. Phys. Chem. C *118*, 7615–7621. https://doi.org/10.1021/jp501237c.

Joule, Volume *8*

Supplemental information

Bifunctional ligand-induced preferred crystal orientation enables highly efficient perovskite solar cells

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Supplementary information

Bifunctional ligand-induced preferred crystal orientation enables highly efficient perovskite solar cells

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Note S1. SCLC analysis of $FAPbI_3$ perovskite films without and with METEAM.

The electron-only device with a structure of glass/FTO/SnO_2/$FAPbI_3$/[6,6]-phenyl-C_{61}-butyric acid methyl ester ($PC_{61}BM$)/Ag was fabricated to determine the trap state density within the perovskite films. The corresponding current-voltage (*I*–*V*) curves are plotted in Figure 3A. The linear correlation of *I*–*V* characteristics at low bias voltage suggests an ohmic response. When the bias voltage exceeds the kink point, the current increases nonlinearly, representing where the trap-states are filled completely, the corresponding bias voltage at the kink point is defined as the trap-filled limit voltage (V_{TFL}), and the trap-state density (n_t) can be determined through Eq. S1:[1]

$$n_t = \frac{2\varepsilon_0\varepsilon_r}{qL^2}\cdot V_{TFL} \tag{S1}$$

where ε_0 is the vacuum permittivity, ε_r is the relative dielectric constant of perovskite (ε_r = 46.9).[2] q is the elementary charge of the electron, and L is the perovskite film thickness (about 750 nm estimated from the cross-sectional SEM images). V_{TFL} is the onset voltage of the trap-filled limit region.

Note S2. OPTP analysis of the $FAPbI_3$ perovskite films without and with METEAM.

The complex photoconductivity $\Delta\tilde{\sigma}$ (ω, τ) was investigated by using the Drude-Smith model,[3,4]

$$\Delta\tilde{\sigma}(\omega,\tau) = \frac{\varepsilon_0\omega_p^2\tau}{1-i\omega\tau}\left[1+\frac{c_1}{1-i\omega\tau}\right] \tag{S2}$$

where the first term $\frac{\varepsilon_0\omega_p^2\tau}{1-i\omega\tau}$ is the Drude conductivity, while the second Smith term $\frac{c_1\Delta\tilde{\sigma}_D}{1-i\omega\tau}$ modifies the Drude model by accounting for the backscattering of carriers due to the influence of disorder in the films. The Drude-Smith model consists of three free parameters: ε_0 is the vacuum permittivity, ω_p, τ and c_1 could be obtained by fitting the experimental data (Figure 3G and 3H). The free charge density could be derived from the Drude plasma frequency ω_p as $n = \frac{\varepsilon_0\omega_p^2 m^*}{e^2}$, where m^* is the effective mass ($m^* = 0.2m$) and e is the elementary charge. The average time interval τ between collision events describes the carrier scattering rate which is related to the carrier mobility ($\mu = \frac{e\tau}{m^*}$). The parameter c_1 describes the strength of backscattering, ranging between -1 (complete backscattering) and 0 (no backscattering). The effective carrier mobility becomes $\mu = \frac{e(1+c_1)\tau}{m^*}$ according to the Drude-Smith model.

Note S3. Calculation of the internal electron-hole quasi-Fermi level splitting (QFLS).

The internal electron-hole quasi-Fermi level splitting (QFLS) from PLQY results were calculated according to the Eq. S7:[5]

$$QFLS = k_BT \times \ln(PLQY \times \frac{J_G}{J_{0,rad}}) \tag{S3}$$

where J_G is the generated current density at 1 sun, it is approximated with the short-circuit current density of the complete solar cell. $J_{0,\ rad}$ is the radiative recombination current in the dark, it is estimated by integrating the overlap of the external quantum efficiency of the full device (EQE) with the black body radiative spectrum at 300 K over the energy.

$$J_{0,rad} = \int EQE\ \emptyset_{BB} d\epsilon \tag{S4}$$

$$\emptyset_{BB} = \frac{1}{4^2h^3c^2}\frac{E^2}{\exp\left(\frac{E}{k_BT}\right)-1} \tag{S5}$$

where E is the photo energy, h is the Plank constant, and c is the light speed in vacuum. The values of $J_{0,\ rad}$ are 1.73 × 10^{-20} and 1.75 × 10^{-20} mA cm^{-2} for the control and target devices respectively according to the EQE spectrum of the full device and the black-body radiative spectrum. The calculated QFLS are shown in Figure S23 and Table S9.

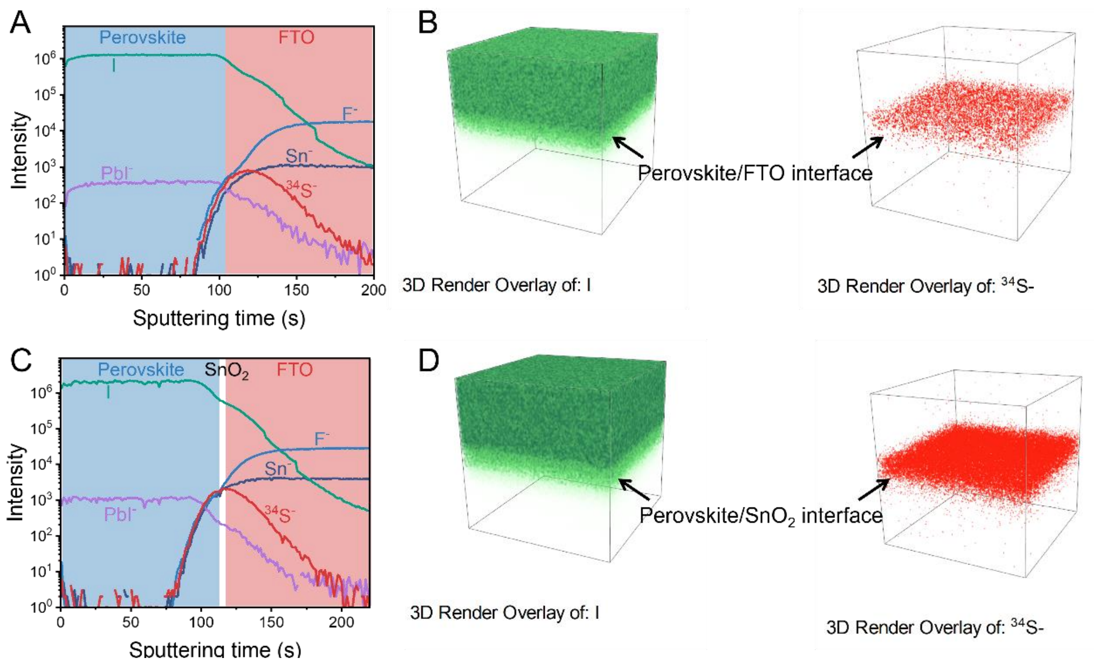

Figure S1. ToF-SIMS curves of the target perovskite films. (A) ToF-SIMS curves of the target perovskite films deposited on the glass/F-doped SnO_2 (FTO) substrate and (B) the corresponding 3D render overlay of I and $^{34}S^-$. (C) ToF-SIMS curves of the target perovskite films deposited on the glass/FTO/SnO_2 substrate (the same figure as Figure 1B) and (D) the corresponding 3D render overlay of I and $^{34}S^-$.

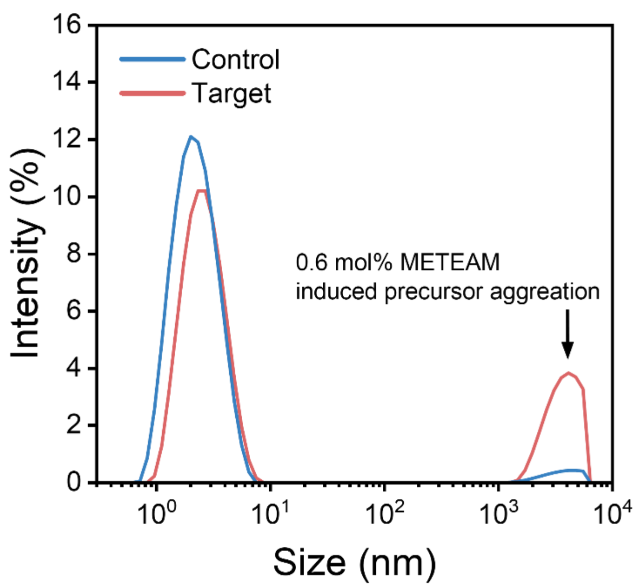

Figure S2. Dynamic light scattering (DLS) spectra of perovskite precursor solutions without and with METEAM ligand. Compared with the control precursor solution, after adding 0.6 mol% METEAM, clearly the proportion of the colloids with large sizes in range of 1–6 μm increases in the target precursor solution, along with decreased proportion of small-sized colloids.

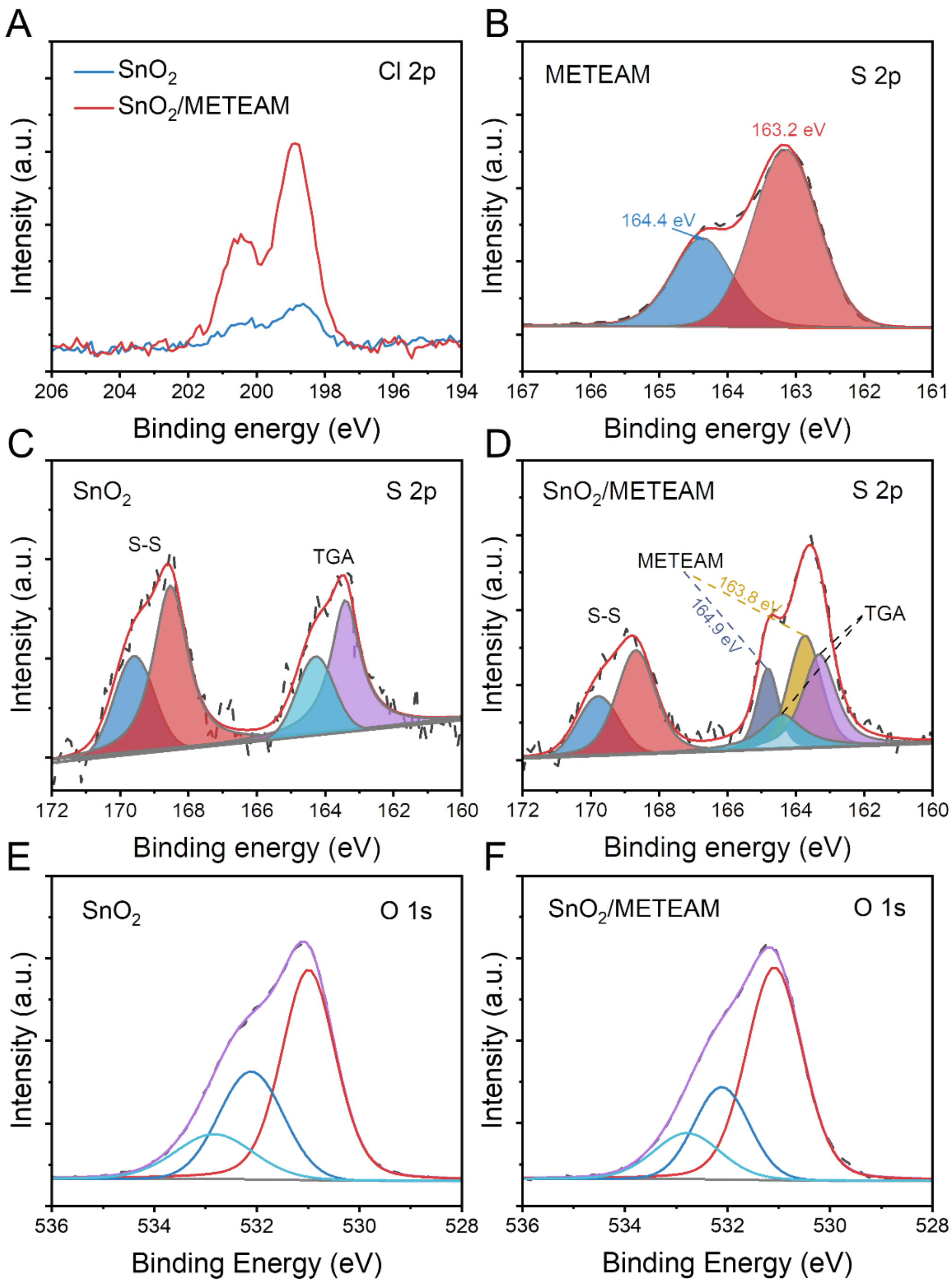

Figure S3. XPS profiles of SnO_2 films without and with METEAM. (A) Cl 2p and (C, D) S 2p high-resolution XPS of SnO_2 before and after METEAM treatment. (B) S 2p high-resolution XPS of METEAM. (E, F) O 1s high-resolution XPS of SnO_2 without and with METEAM treatment. C 1s spectrum was used to calibrate all XPS spectra. The SnO_2/METEAM sample for XPS measurement was prepared by spin-coating the METEAM solution dissolved in isopropyl alcohol (concentration: 1 mg/mL) onto the SnO_2 substrate and then annealed at 100 °C for 10 minutes. In Figure S3C, two S 2p signals were detected for the pristine SnO_2, which were assigned to the residual $HOOCCH_2SH$ (thioglycolic acid, TGA) and $HOOCCH_2SSCH_2COOH$ (abbreviated as S-S).[6] O 1s high-resolution XPS could be deconvoluted to three peaks which could be classified as the lattice oxygen (~531.0 eV, O_{lat}), the oxygen vacancy (~532.1 eV, O_V), and hydroxyl adsorbed onto SnO_2 (~532.8 eV, O_{OH}).[7] After METEAM treatment, the ratio of oxygen vacancy decreases from 30% to 24% as listed in Table S1.

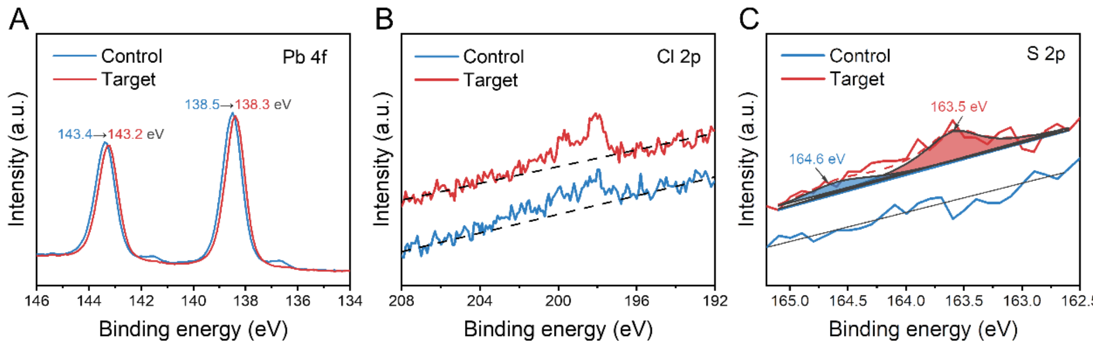

Figure S4. XPS profiles of $FAPbI_3$ perovskite films without and with METEAM. (A) Pb 4f, (B) Cl 2p, and (C) S 2p high-resolution XPS spectra of $FAPbI_3$ perovskite films without and with METEAM. C 1s spectrum was used to calibrate all XPS spectra. The Cl signals stem from the not fully sublimated MACl, which has been demonstrated by the previous works.[8]

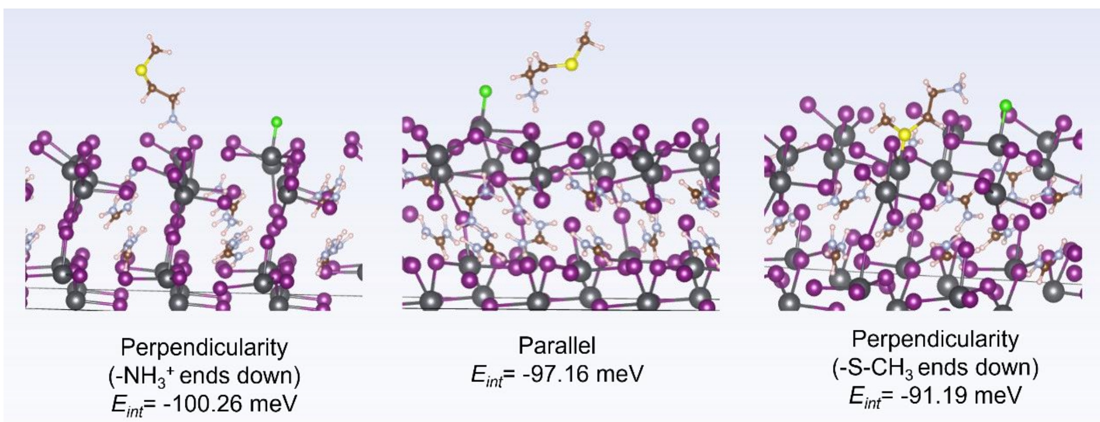

Figure S5. The various equilibrium configurations of the METEAM molecular interaction with (100) facets of α-$FAPbI_3$ perovskite. $METEAM^+$ with $-S-CH_3$ group exposed to the Pb–I framework shows E_{int} of -91.19 meV, which is lower than that of $METEAM^+$ with $-H_3N^+$ group (E_{int} = -100.26 meV) exposed to the Pb–I framework, resulting in a spontaneous configuration transition from the former to the latter.

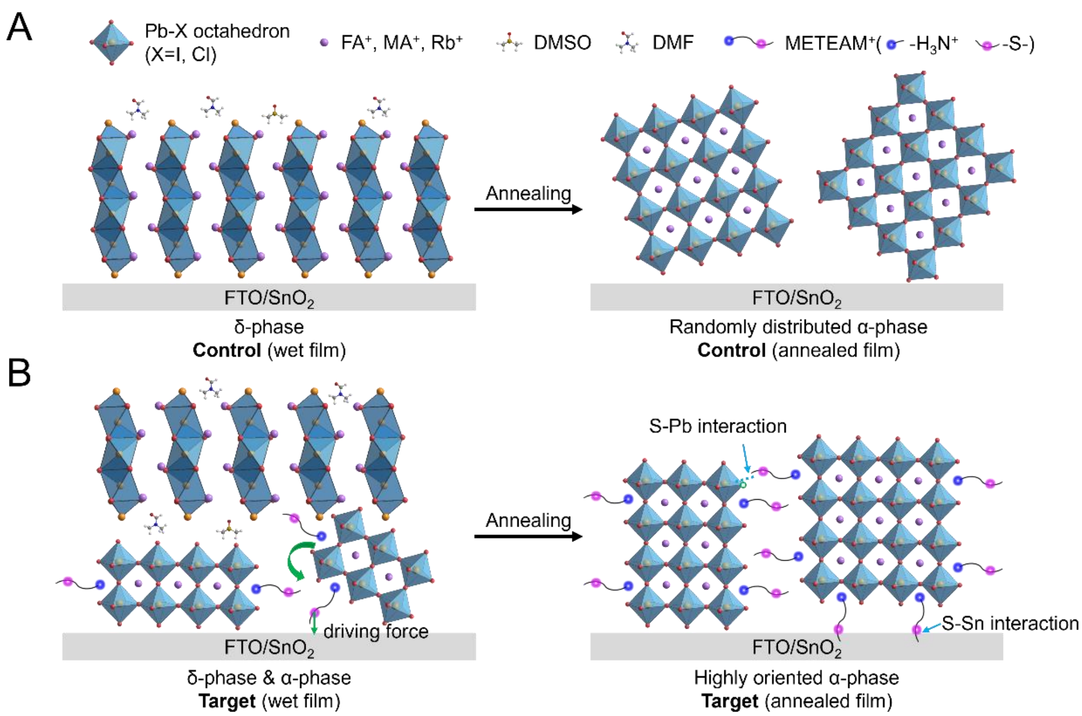

Figure S6. Schematic diagram of the crystal growth processes of the control and target perovskite films. METEAM lowers the formation energy of α-phase $FAPbI_3$ forming a randomly distributed α-phase before thermal annealing, and the S-Sn interaction generates the driving force to deflect the crystal with (100) plane parallel to the substrate. Then, the well-arranged α-phase $FAPbI_3$ acts as templates to trigger the transition of δ-phase to α-phase, resulting in preferred orientation finally.

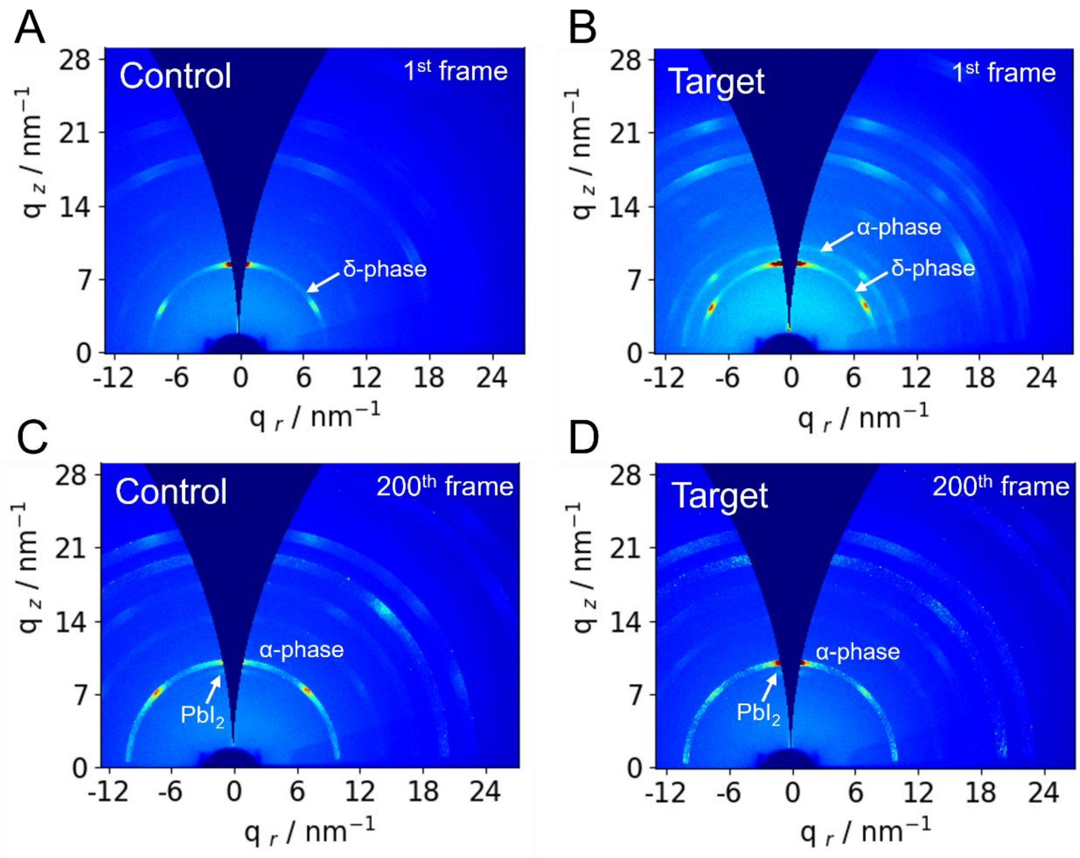

Figure S7. GIXRD profiles of the $FAPbI_3$ perovskite films without and with METEAM. GIXRD patterns of $FAPbI_3$ films without (control) and with (target) METEAM. The GIXRD patterns were recorded every five seconds. In the initial stage (Figures 2A and 2B, Figures S7A and S7B), no PbI_2 diffraction signal was detected although the molar ratio of PbI_2 and FAI is larger than 1 in the perovskite precursor solution. After annealing at 120 °C for about 1000 seconds (200th frame), PbI_2 diffraction signals appeared. Very likely MACl additive participates in the formation of the perovskite phase and sublimates during the thermal annealing process, leading to the residual of PbI_2.[9]

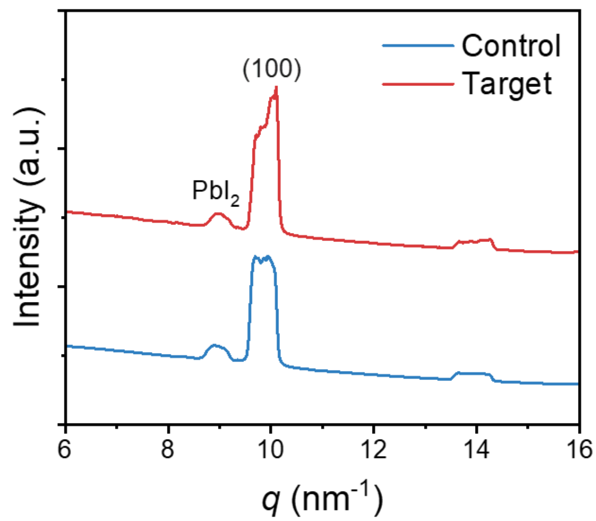

Figure S8. Azimuth integral curves of the $FAPbI_3$ perovskite films without and with METEAM. Azimuth integral curves for the (100) plane extracted from the GIXRD patterns in Figures 2H and 2I.

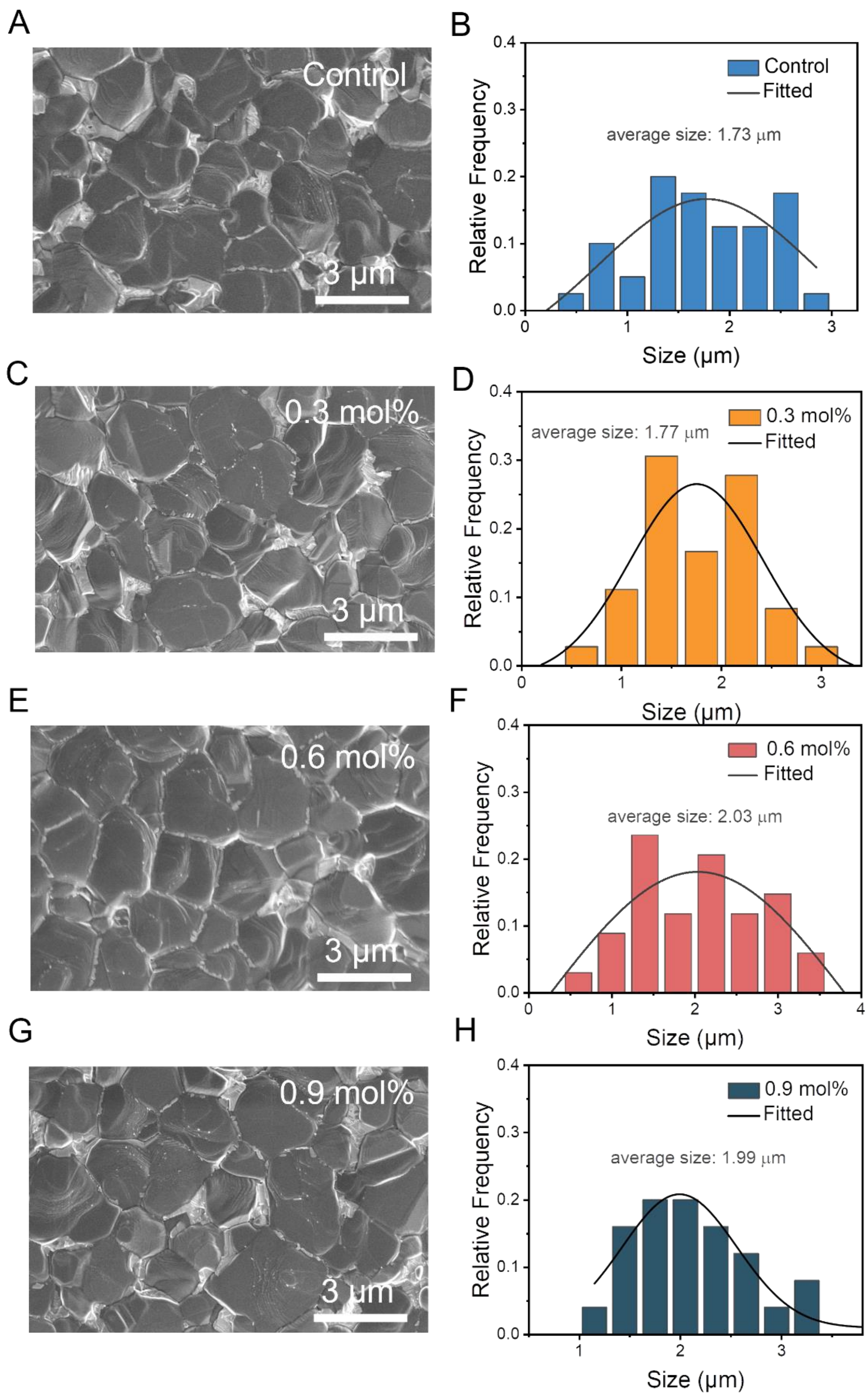

Figure S9. Top-surface SEM images of the control and target $FAPbI_3$ perovskite films with different METEAM concentrations (left) and the corresponding grain size statistics (right) extracted from the top-view SEM images.

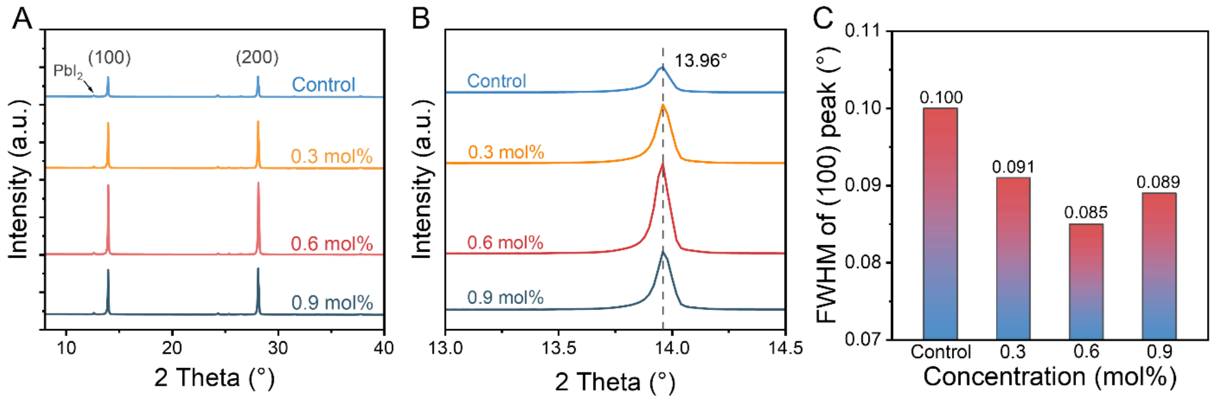

Figure S10. XRD patterns of the control and target $FAPbI_3$ perovskite films with different METEAM concentrations. (A) XRD patterns, (B) Enlarged XRD patterns, and (C) FWHM of (100) plane.

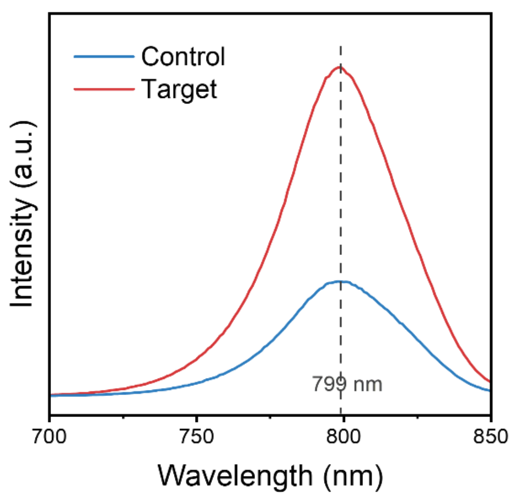

Figure S11. PL spectra of $FAPbI_3$ perovskite films on perovskite/glass substrates. Steady state PL spectra of the $FAPbI_3$ perovskite films with and without METEAM.

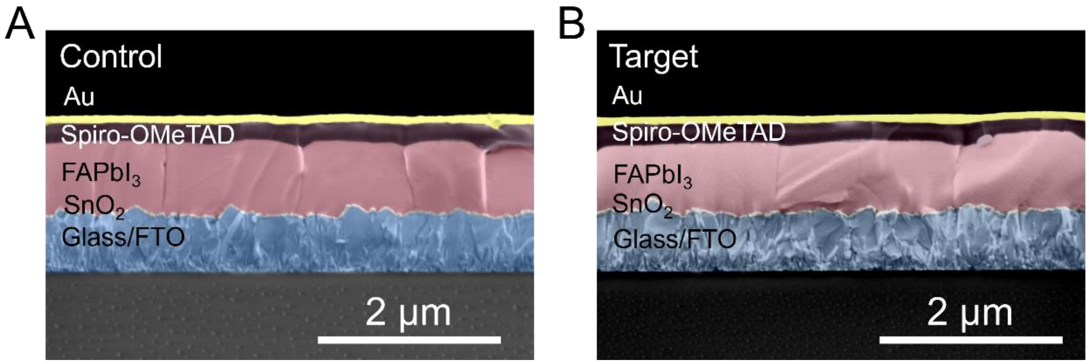

Figure S12. Cross-sectional SEM images of the control and target $FAPbI_3$ perovskite films along with other layers.

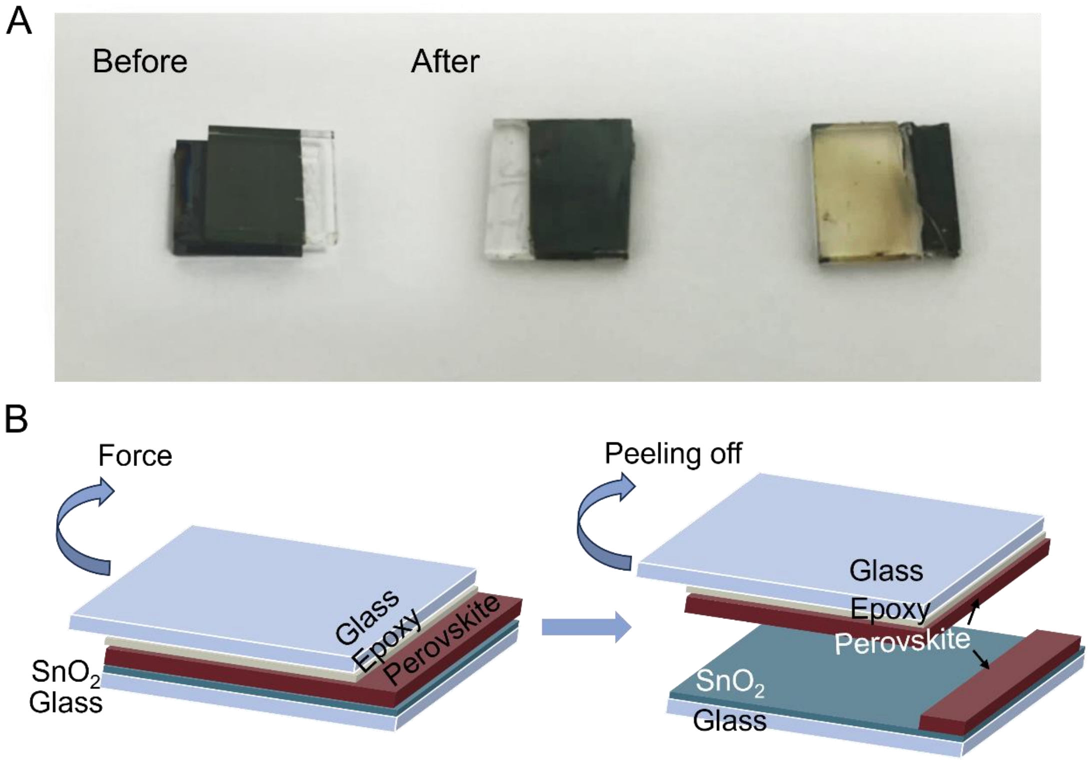

Figure S13. Photos of the perovskite films deposited on SnO_2/FTO substrate and schematic diagram of exposing the buried interface. (A) The photography of perovskite films before and after peeling off. The top surface of the as-prepared perovskite film was partly pasted by a glass slide through UV glue. After the glue was solidified, the film was peeled off from the SnO_2/FTO substrate, then the bottom surface (buried interface) of the perovskite film is exposed for characterization. (B) Schematic illustration of peeling off perovskite films from SnO_2/FTO substrates.

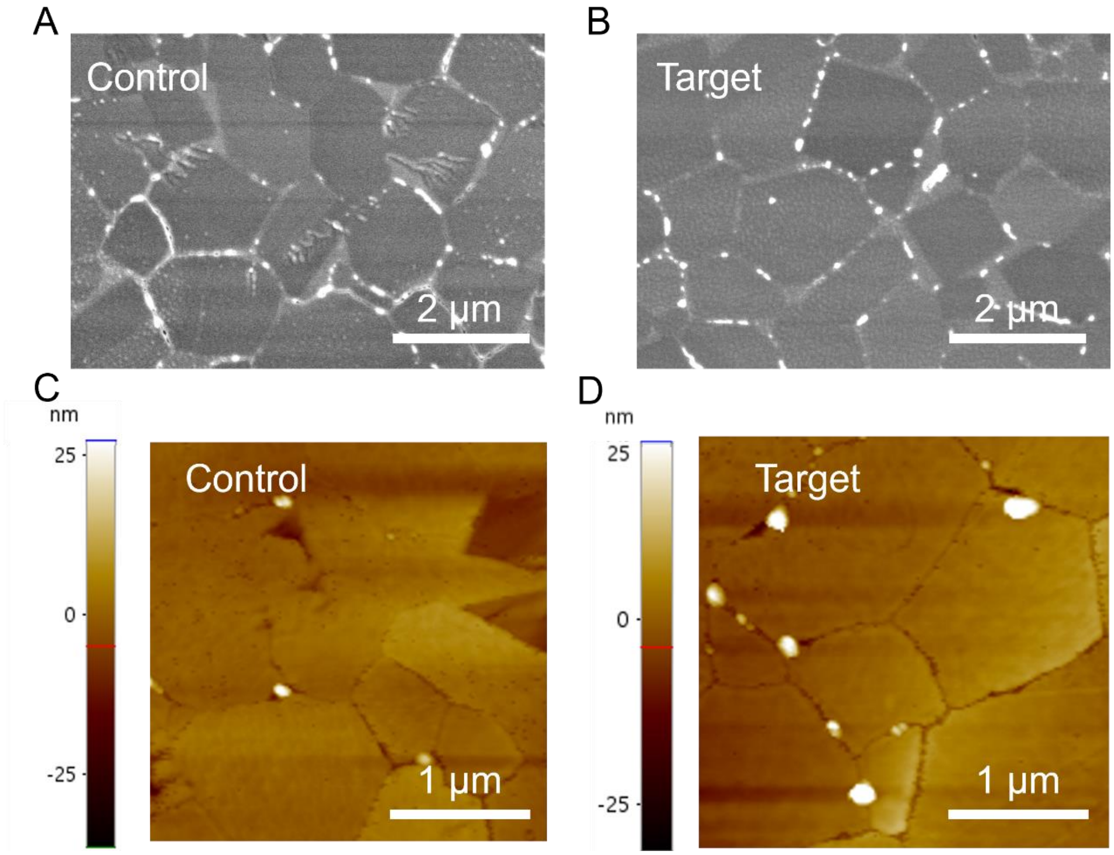

Figure S14. The SEM and AFM images on the buried interface of the control and target perovskite layer exfoliated from SnO_2/FTO substrate. Top-view SEM images (A, B) and AFM images (3×3 μm^2, C, D) of the buried interface observed from the bottom of perovskite films peeled off from the SnO_2/FTO substrate.

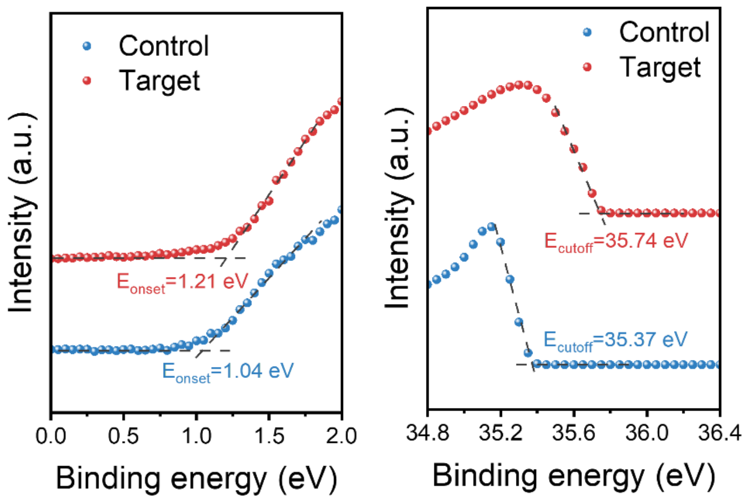

Figure S15. **UPS curves of $FAPbI_3$ perovskite films without and with METEAM.** Corresponding energy level diagrams of the $FAPbI_3$ perovskite films without and with METEAM additive. Ultraviolet photoelectron spectroscopy (UPS) is usually employed to determine the energy band structure of perovskite films with respect to the vacuum level (E_{VAC} = 0 eV). The work function (WF) was calculated by WF = $\hbar\nu - E_{\mathrm{cutoff}}$ from the secondary electron threshold, where the excitation photon energy ($\hbar\nu$) is 40 eV. The valence band maximum is found by a linear extrapolation of the leading edge of the Fermi edge of the spectra. The valence band (E_V) is calculated by E_V = - (WF + E_{onset}). Accordingly, we further calculated the conduction band energy level (E_C) by $E_C = E_V + E_g$ (band gap of the perovskite), where band gaps are extracted from the optical absorption spectrum (Figure S16) via a Tauc Plot method in which extending the section of the straight line of the curve to the x-axis and the intercept of the line gives the optical band gap. The binding energy (E_b) is converted by $E_b = \hbar\nu - \mathrm{WF}\ (Instrument) - V_{bi} - E_k$, where the work function of the instrument is 4.3 eV, the applied bias to the sample V_{bi} is -5 V, E_k is the kinetic energy measured directly by the instrument. The related parameters were summarized in Table S4.

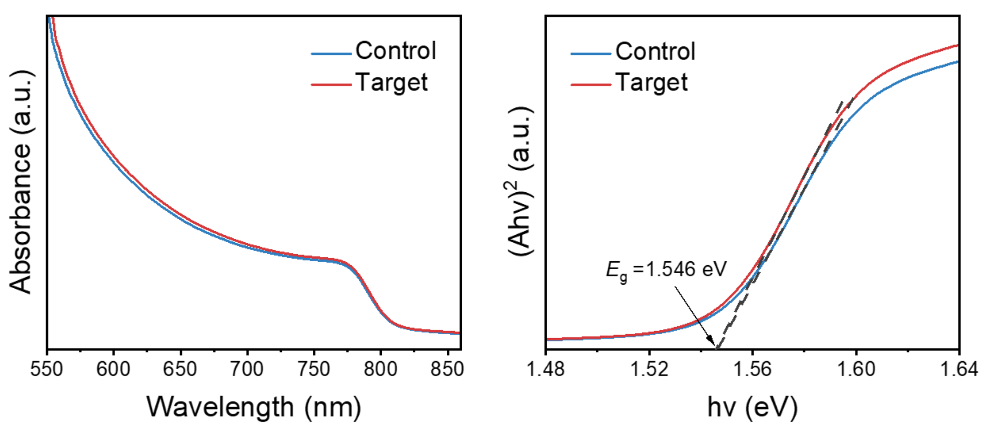

Figure S16. **Ultraviolet photoelectron spectroscopy and optical characteristics.** The optical absorption spectrum and Tauc plots of the $FAPbI_3$ perovskite without (control) and with (target) METEAM.

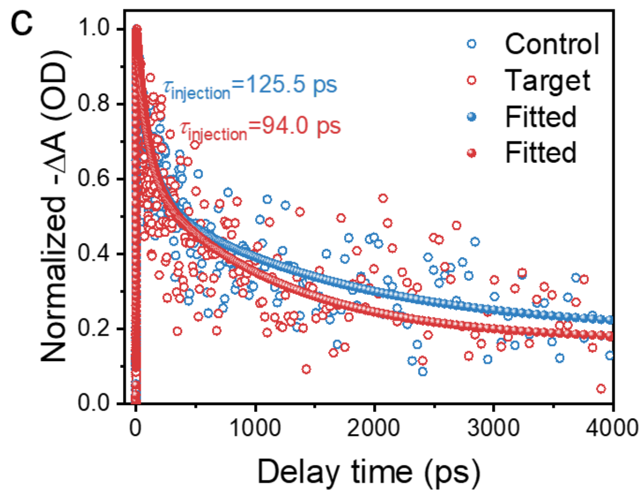

Figure S17. TAS curves of $FAPbI_3$ perovskite films without and with METEAM. Normalized decay kinetic curves at 778 nm of the TAS curves of the control and target perovskite films.

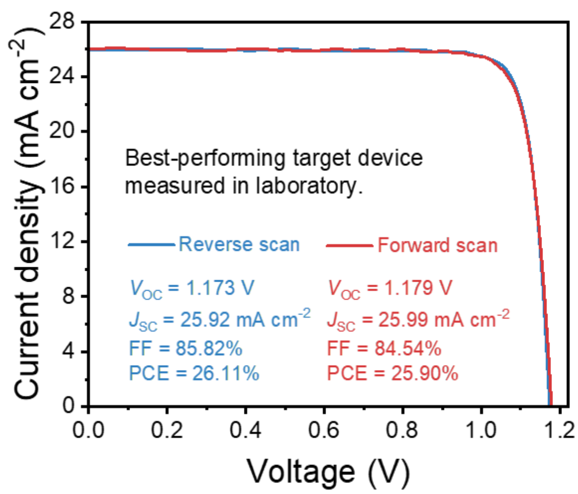

Figure S18. Best photovoltaic performance device measured in the laboratory. The *J*-*V* curves in reverse and forward scans of the best-performing target device measured in the laboratory.

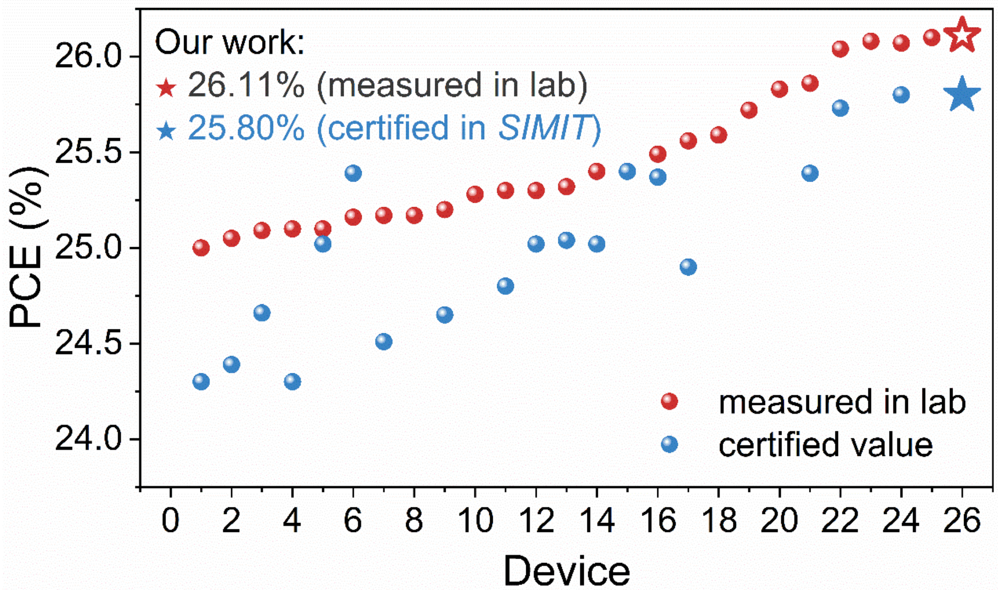

Figure S19. The comparison between PCE measured in laboratory and certified PCE. Included the recently reported works with PCE exceeding 25%.

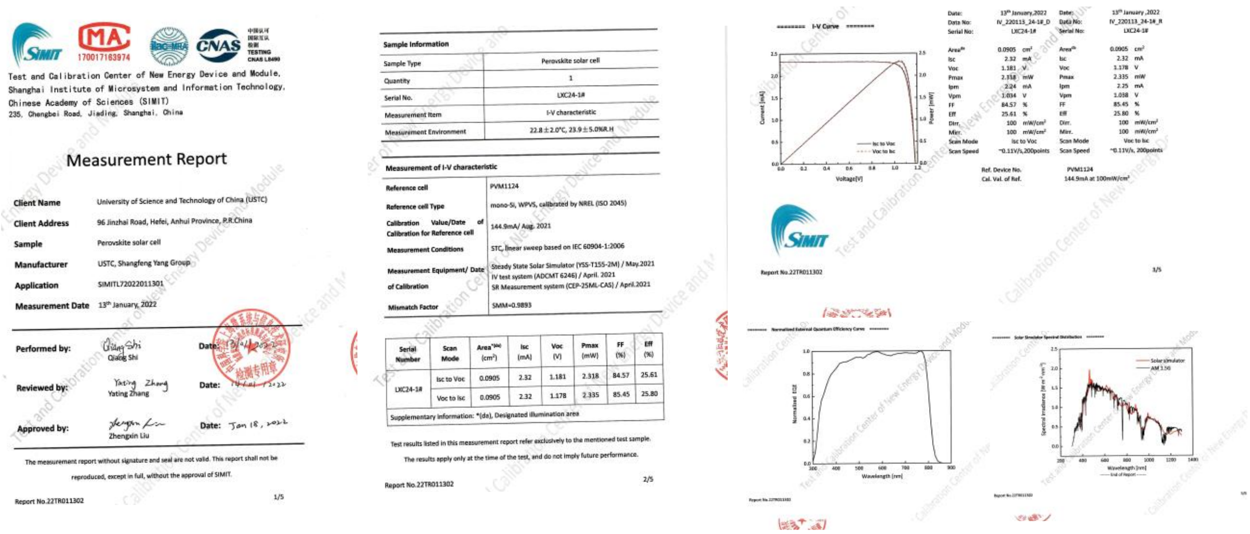

Figure S20. Certified data for the best-performing device. The certified result for the best-performing device based on $FAPbI_3$ with METEAM additive measured in Shanghai Institute of Microsystem and Information Technology (*SIMIT*).

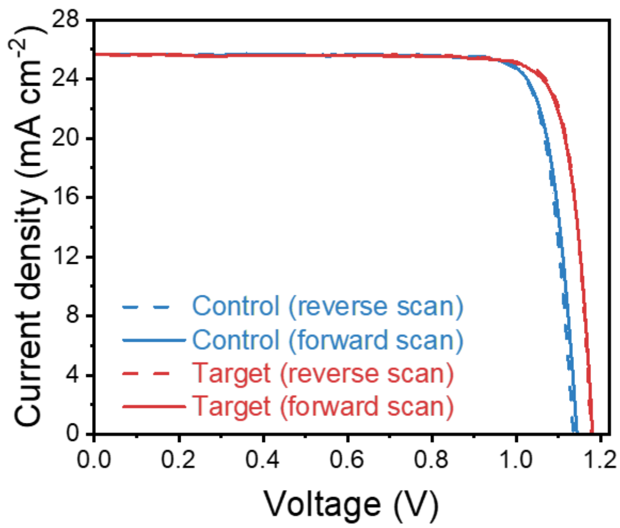

Figure S21. The *J-V* curves of the device are measured in *SIMIT*. The *J-V* curves in reverse and forward scans of the best-performing devices based on $FAPbI_3$ without (control) and with (target) METEAM additive measured in *SIMIT*.

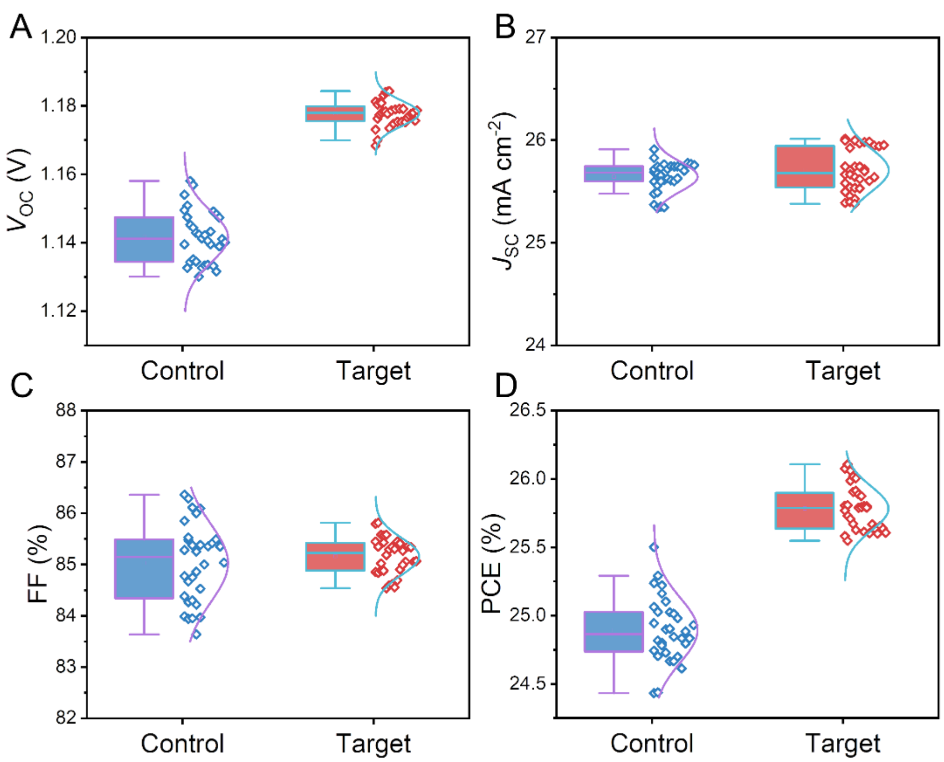

Figure S22. **Device statistics for the control and target devices.** (A) V_{OC}, (B) J_{SC}, and (C) FF and (D) PCE

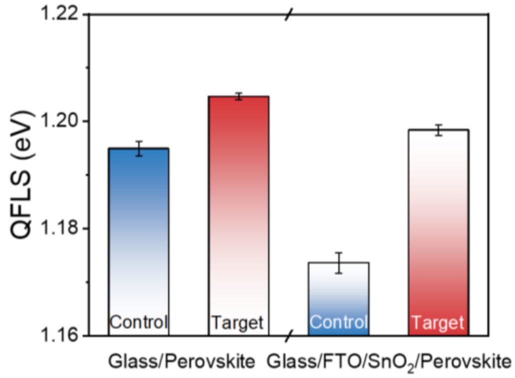

Figure S23. Calculation of the internal electron-hole quasi-Fermi level splitting (QFLS). QFLS for the $FAPbI_3$ perovskite without and with METEAM deposited on the glass and glass/FTO/SnO_2 substrate.

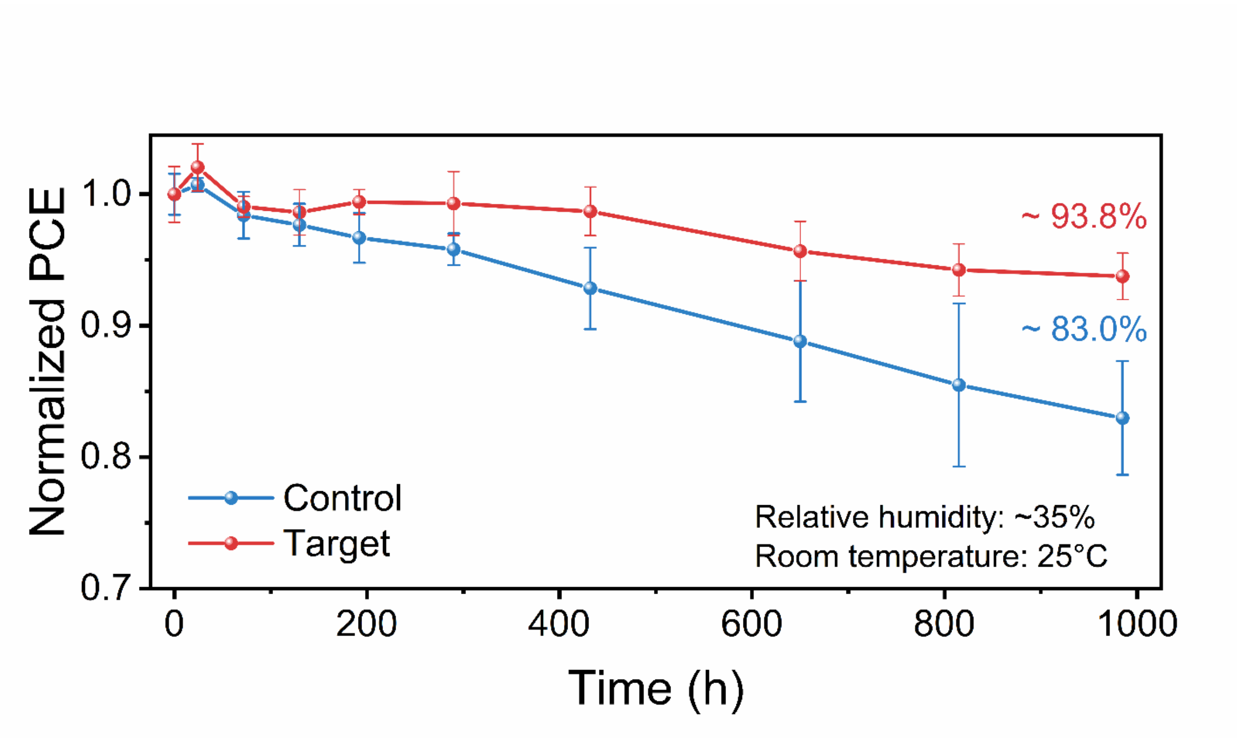

Figure S24. Ambient stability measurement of the devices without and with METEAM. The evolution of normalized PCE of the unencapsulated devices under the ambient environment with a relative humidity of ~35% and temperature of 25°C. The enhanced ambient environment storage stability is ascribed to the reduced defects stemming from the passivation function of METEAM at the buried interface.

Table S1. The percentage of oxygen species of SnO_2 obtained from the O 1s XPS peak area integration.

Sample	Position (eV)	Assignment	Area	Ratio	Total ratio
SnO_2	531.0	O_{lat}	75475	1	0.54
	532.1	O_v	40820	0.54	0.30
	532.8	O_{OH}	21979	0.29	0.16
SnO_2/METEAM	531.0	O_{lat}	77586	1	0.61
	532.1	O_v	30652	0.40	0.24
	532.8	O_{OH}	19503	0.25	0.15

Table S2. Interaction energies of $METEAM^+$ with PbI_2 exposure of different crystal planes.

Crystal planes	Interaction energy (eV)	Per atom interaction energy (E_{int}, meV)
(100)	-9.39	-100.26
(110)	-4.98	-53.18
(111)	-7.27	-77.62

Table S3. Fitted parameters of the time-resolved PL spectra of the $FAPbI_3$ perovskite without (control) and with (target) METEAM.

	A_1	τ_1 (ns)	A_2	τ_2 (ns)	τ_{ave} (ns)
Control	0.663	168.6	0.269	1385.4	1104.6
Target	0.582	361.0	0.319	1813.0	1426.1

The results are obtained by using the double exponential equation (Eq. S6) to fit the time-resolved PL spectra and τ_{ave} is calculated according to the Eq. S7.

$$f(t) = A_1\exp(-\frac{t}{\tau_1}) + A_2\exp(-\frac{t}{\tau_2}) + B \quad \text{(S6)}$$

$$\tau_{ave} = \frac{A_1\tau_1^2+A_2\tau_2^2}{A_1\tau_1 + A_2\tau_2} \quad \text{(S7)}$$

Table S4. Detailed parameters calculated from the UPS and optical absorption spectrum of the $FAPbI_3$ perovskite without (control) and with (target) METEAM.

Sample	E_F (eV)	WF (eV)	E_{onset} (eV)	E_g (eV)	E_V (eV)	E_C (eV)
Control	-4.63	4.63	1.04	1.55	-5.67	-4.12
Target	-4.26	4.26	1.21	1.55	-5.47	-3.92

Table S5. Fitted kinetics parameters of transient absorption spectra of the control and target perovskite films deposited on glass/FTO/SnO_2/substrate measured at 778 nm.

	A_1	τ_1 (ps)	A_2	τ_2 (ps)	τ_{ave} (ps)
Control	0.49	125.5	0.51	1644	1540
Target	0.48	94.0	0.52	1151	1077

The results are fitted with the following double exponential equation.[11]

$$y = y_0 + A_1\exp(-\frac{t}{\tau_1}) + A_2\exp(-\frac{t}{\tau_2}) \quad \text{(S8)}$$

The average decay time (τ_{ave}) was calculated through the following equation.

$$\tau_{ave} = \frac{A_1\tau_1^2 + A_2\tau_2^2}{A_1\tau_1 + A_2\tau_2} \quad \text{(S9)}$$

The longer decay time (τ_2) represents the excited-state decay or carrier recombination in perovskite films, while the shorter decay time (τ_1) reflects the electron injection from photoexcited $FAPbI_3$ to the E_C of SnO_2 ETL.[11]

Table S6. The photovoltaic parameters of the control and target devices with different METEAM concentrations.

Concentration	V_{OC} [a] (V)	J_{SC} [a] (mA cm^{-2})	FF [a] (%)	PCE [a] (%)
Control	1.141±0.005	25.62±0.13	85.18±0.66	24.91±0.16
0.3 mol%	1.149±0.003	25.81±0.12	85.17±1.20	25.26±0.28
0.6 mol%	1.176±0.004	25.73±0.20	85.29±0.32	25.81±0.19
0.9 mol%	1.184±0.004	25.45±0.07	84.29±0.33	25.40±0.08

[a] Averaged from 15 devices fabricated independently.

Table S7. Photovoltaic parameters of the control and target devices measured under one sun illumination (AM 1.5G, 100 mA cm^{-2}).

Device		V_{OC} (V)	J_{SC} (mA cm^{-2})	FF (%)	PCE (%)
Control	Average [a]	1.142±0.007	25.64±0.14	85.03±0.75	24.89±0.23
	Best	1.154	25.91	85.28	25.50
Target	Average [a]	1.178±0.004	25.71±0.21	85.16±0.36	25.78±0.16
	Best	1.173	25.92	85.82	26.11

[a] Averaged 32 devices fabricated independently.

Table S8. Detailed PCE values plotted in Figure S19.

Device structure	Champion PCE (Lab.)	Certified PCE		Device
		J-V scan	QSS	
ITO/PTAA/$(FA_{0.98}MA_{0.02})_{0.95}Cs_{0.05}Pb(I_{0.95}Br_{0.02})_3$/$C_{60}$/BCP/Ag	25.0%	24.3%	—	1 (ref. 1)
FTO/SnO_2/$Cs_{0.05}(FA_{0.92}MA_{0.08})_{0.95}Pb(I_{0.92}Br_{0.08})_3$/spiro-OMeTAD/Ag	25.05%	24.39%	—	2 (ref. 12)
FTO/SnO_2/1.0 mol% $CsPbBr_3$-$FAPbI_3$/spiro-OMeTAD/Au	25.09%	24.66%	—	3 (ref. 2)
FTO/SnO_2/$FA_{1-x}MA_xPbI_{3-y}Br_y$/spiro-OMeTAD/Au	25.1%	24.30%	—	4 (ref. 13)

FTO/c-TiO_2/m-TiO_2/$FAPbI_3$/spiro-OMeTAD/Au	25.1%	25.02%	—	5 (ref. 14)
ITO/MPA-CPA/$Cs_{0.05}(FA_{0.95}MA_{0.05})_{0.95}Pb(I_{0.95}Br_{0.05})_3$/$C_{60}$/BCP/Ag	25.16%	25.39%	—	6 (ref. 5)
FTO/SnO_2/0.8 mol% $MAPbBr_3$-$FAPbI_3$/spiro-OMeTAD/MoO_3/Ag	25.17%	24.51%	—	7 (ref. 15)
FTO/c-TiO_2/m-TiO_2/$(FAPbI_3)_{1-x}(MDA, Cs)_x$/spiro-OMeTAD/Au	25.17%	—	24.37%	8 (ref. 16)
ITO/SnO_2/$FA_{1-x}Cs_yPbI_{3-y}Cl_y$/spiro-OMeTAD/Au	25.2%	24.65%	—	9 (ref. 17)
FTO/c-TiO_2/m-TiO_2/$FAPbI_3$/spiro-OMeTAD/Au	25.28%	—	24.68%	10 (ref. 18)
FTO/c-TiO_2/$FAPbI_3$/spiro-OMeTAD/Au	25.3%	24.8%	—	11 (ref. 19)
FTO/SnO_2/$FAPbI_3$/spiro-OMeTAD/Au	25.30%	25.02%	—	12 (ref. 20)
FTO/c-TiO_2/SnO_2/$Cs_{0.05}MA_{0.05}FA_{0.9}PbI_3$/spiro-OMeTAD/Au	25.32%	25.04%	—	13 (ref. 21)
FTO/c-TiO_2/SnO_2/$Cs_{0.05}MA_{0.05}FA_{0.9}PbI_3$/spiro-OMeTAD/Au	25.4%	25.02%	—	14 (ref. 22)
FTO/SnO_2/0.8 mol% $MAPbBr_3$-$FAPbI_3$/spiro-OMeTAD/Au	—	25.4%	25.17%	15 (ref. 23)
ITO/Me-2PACZ/$Rb_{0.05}Cs_{0.05}MA_{0.05}FA_{0.85}Pb(I_{0.95}Br_{0.05})_3$/LiF/$C_{60}$/BCP/Ag	25.49%	25.37%	24.05%	16 (ref. 24)
ITO/Me-4PACZ/PIC/$(FA_{0.95}MA_{0.05})_{0.95}Cs_{0.05}Pb(I_{0.95}Br_{0.05})_3$/LiF/$C_{60}$/BCP/Ag	25.56%	24.90%	—	17 (ref. 25)
FTO/c-TiO_2/m-TiO_2/2% Fo-$FAPbI_3$/spiro-OMeTAD/Au	25.59%	—	25.21%	18 (ref. 26)
FTO/paa-QD-SnO_2@c-TiO_2 $FAPbI_3$/spiro-OMeTAD/Au	25.72%	—	25.39%	19 (ref. 27)
FTO/SnO_2/$FAPbI_3$:$MDACl_2$/spiro-OMeTAD/Au	25.83%	—	25.49%	20 (ref. 28)
ITO/$FA_{1-x-y}MA_xCs_yPbI_{3-z}Br_z$ (DMAcPA)/$PC_{61}BM$/BCP/Ag	25.86%	25.39%	—	21 (ref. 29)
FTO/SnO_2/$MAPbBr_3$-$FAPbI_3$/spiro-OMeTAD/Au	26.04%	25.73%	25.06%	22 (ref. 30)

FTO/SnO_2/$FAPbI_3$:$MDACl_2$/spiro-OMeTAD/Au	26.08%	—	25.73%	23 (ref. 31)
FTO/SnO_2/$FAPbI_3$/spiro-OMeTAD/Au	26.07%	25.80%		24 (ref. 32)
FTO/SnO_2/$FA_{1-x}Rb_xPbI_3$/spiro-OMeTAD/Au	26.10%	—	25.56%	25 (ref. 8)
FTO/SnO_2/$FA_{1-x}Rb_xPbI_3$(METEAM)/spiro-OMeTAD/Au	**26.11%**	**25.80%**	**—**	**This work**

Table S9. Detailed photovoltaic parameters of the *J-V* curves in reverse and forward scans the best-performing devices based on $FAPbI_3$ without (control) and with (target) METEAM additive measured in *SIMIT*.

Devices	Scan direction	V_{OC} (V)	J_{SC} (mA cm^{-2})	*FF* (%)	PCE (%)	HI[a] (%)
Control	Reverse	1.136	25.67	85.10	24.81	0.12
	Forward	1.144	25.70	84.26	24.78	
Target	Reverse	1.178	25.64	85.45	25.80	0.74
	Forward	1.182	25.63	84.57	25.61	

[a] HI: hysteresis index. HI = ($PCE_{Reverse}$ - $PCE_{Forward}$)/$PCE_{Reverse}$

Table S10. PLQY and QFLS for the perovskite without and with METEAM deposited on the glass and glass/FTO/SnO_2 substrate.

Sample	Glass/Perovskite		Glass/FTO/SnO_2/Perovskite	
	Control	Target	Control	Target
PLQY (%)	8.02±0.42	11.78±0.28	3.52±0.26	9.24±0.36
QFLS (eV)	1.195±0.001	1.205±0.001	1.174±0.002	1.198±0.001

Note: Averaged 5 samples fabricated independently.

References

1. Li, Z., Li, B., Wu, X., Sheppard, S. A., Zhang, S. F., Gao, D. P., Long, N. J., Zhu, Z. L. (2022). Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells. Science *376*, 416-420.
2. Li, X. C., Wu, X., Li, B., Cen, Z. Y., Shang, Y. B., Lian, W. T., Cao, R., Jia, L. B., Li, Z., Gao, D. P., et al. (2022). Modulating the deep-level defects and charge extraction for efficient perovskite solar cells with high fill factor over 86%. Energy Environ. Sci. *15*, 4813-4822.
3. La-o-vorakiat, C., Salim, T., Kadro, J., Khuc, M. T., Haselsberger, R., Cheng, L., Xia, H. X., Gurzadyan, G. G., Su, H. B., Lam, Y. M., et al. (2015). Elucidating the role of disorder and free-carrier recombination kinetics in $CH_3NH_3PbI_3$ perovskite films. Nat. Commun. *6*, 7903.
4. Lin, R. X., Xu, J., Wei, M. Y., Wang, Y. R., Qin, Z. Y., Liu, Z., Wu, J. L., Xiao, K., Chen, B., Park, S. M., et al. (2022). All-perovskite tandem solar cells with improved grain surface passivation. Nature *603*, 73-78.
5. Zhang, S., Ye, F. Y., Wang, X. Y., Chen, R., Zhang, H. D., Zhan, L. Q., Jiang, X. Y., Li, Y. W., Ji, X. Y., Liu, S. J., et al. (2023). Minimizing buried interfacial defects for efficient inverted perovskite solar cells. Science *380*, 404-409.
6. Zhao, Q. Q., Liu, D. C., Li, Z. P., Zhang, B. Q., Sun, X. H., Shao, Z. P., Chen, C., Wang, X., Hao, L. Z., Wang, X. Z., et al. (2022). Chemical bath deposition of mesoporous SnO_2 to improve interface adhesion and device operational stability. Chem. Eng. J. *443,* 136308.
7. Liu, J. L., Li, S., Liu, S., Chu, Y. M., Ye, T., Qiu, C., Qiu, Z. X., Wang, X. D., Wang, Y. F., Su, Y. Q., et al. (2022). Oxygen Vacancy Management for High-Temperature Mesoporous SnO_2 Electron Transport Layers in Printable Perovskite Solar Cells. Angew. Chem. Int. Edit. *61,* e2022020.
8. Zhao, Y., Ma, F., Qu, Z. H., Yu, S. Q., Shen, T., Deng, H. X., Chu, X. B., Peng, X. X., Yuan, Y. B., Zhang, X. W., et al. (2022). Inactive $(PbI_2)_{(2)}RbCl$ stabilizes perovskite films for efficient solar cells. Science *377*, 531-534.
9. Kim, M., Kim, G. H., Lee, T. K., Choi, I. W., Choi, H. W., Jo, Y., Yoon, Y. J., Kim, J. W., Lee, J., Huh, D., et al. (2019). Methylammonium Chloride Induces Intermediate Phase Stabilization for Efficient Perovskite Solar Cells. Joule *3*, 2179-2192.
10. Xu, J. X., Boyd, C. C., Yu, Z. S. J., Palmstrom, A. F., Witter, D. J., Larson, B. W., France, R. M., Werner, J., Harvey, S. P., Wolf, E. J., et al. (2020). Triple-halide wide-band gap perovskites with suppressed phase segregation for efficient tandems. Science *367*, 1097-1104.
11. Zhang, J., Wang, L., Jiang, C., Cheng, B., Chen, T., Yu, J. (2021). $CsPbBr_3$ nanocrystal induced bilateral interface modification for efficient planar perovskite solar cells. Adv. Sci. *8*, 2102648.
12. Luo, X. H., Shen, Z. C., Shen, Y. Z., Su, Z. H., Gao, X. Y., Wang, Y. B., Han, Q. F., Han, L. Y. (2022). Effective Passivation with Self-Organized Molecules for Perovskite Photovoltaics. Adv. Mater. *34*, 202202100.
13. Shen, Z. C., Han, Q. F., Luo, X. H., Shen, Y. Z., Wang, T., Zhang, C. Y., Wang, Y. B., Chen, H., Yang, X. D., Zhang, Y. Q., et al. (2022). Crystal-array-assisted growth of a perovskite absorption layer for efficient and stable solar cells. Energy Environ. Sci. *15*, 1078-1085.
14. Zhang, T. K., Wang, F., Kim, H. B., Choi, I. W., Wang, C. F., Cho, E., Konefal, R., Puttisong, Y., Terado, K., Kobera, L., et al. (2022). Ion-modulated radical doping of spiro-OMeTAD for more efficient and stable perovskite solar cells. Science *377*, 495-501.
15. Wu, Y., Wang, Q., Chen, Y., Qiu, W., Peng, Q. (2022). Stable perovskite solar cells with 25.17% efficiency enabled by improving crystallization and passivating defects synergistically. Energy Environ. Sci. *15*, 4700-4709.
16. Kim, G., Min, H., Lee, K. S., Lee, D. Y., Yoon, S. M., Seok, S. I. (2020). Impact of strain relaxation on performance of alpha-formamidinium lead iodide perovskite solar cells. Science *370*, 108-112.
17. Luo, C., Zheng, G., Wang, X., Gao, F., Zhan, C., Gao, X., Zhao, Q. (2023). Solid–solid chemical bonding featuring targeted defect passivation for efficient perovskite photovoltaics. Energy Environ. Sci. *16*, 178-189.
18. Kim, M., Choi, I. W., Choi, S. J., Song, J. W., Mo, S. I., An, J. H., Jo, Y., Ahn, S., Ahn, S. K., Kim, G. H., et al. (2021). Enhanced electrical properties of Li-salts doped mesoporous TiO2 in perovskite solar cells. Joule *5*, 659-672.
19. Yang, T., Gao, L., Lu, J., Ma, C., Du, Y., Wang, P., Ding, Z., Wang, S., Xu, P., Liu, D., et al. (2023). One-stone-for-two-birds strategy to attain beyond 25% perovskite solar cells. Nat. Commun. *14*, 839.
20. Wu, J., Li, M. H., Fan, J. T., Li, Z., Fan, X. H., Xue, D. J., Hu, J. S. (2023). Regioselective Multisite

Atomic-Chlorine Passivation Enables Efficient and Stable Perovskite Solar Cells. J. Am. Chem. Soc. *145*, 5872-5879.

21. Yang, W., Ding, B., Lin, Z., Sun, J., Meng, Y., Ding, Y., Sheng, J., Yang, Z., Ye, J., Dyson, P. J., et al. (2023). Visualizing Interfacial Energy Offset and Defects in Efficient 2D/3D Heterojunction Perovskite Solar Cells and Modules. Adv. Mater. *35,* 2302071.
22. Shi, P., Ding, Y., Ding, B., Xing, Q., Kodalle, T., Sutter-Fella, C. M., Yavuz, I., Yao, C., Fan, W., Xu, J., et al. (2023). Oriented nucleation in formamidinium perovskite for photovoltaics. Nature *620*, 323–327.
23. Yoo, J. J., Seo, G., Chua, M. R., Park, T. G., Lu, Y. L., Rotermund, F., Kim, Y. K., Moon, C. S., Jeon, N. J., Correa-Baena, J. P., et al. (2021). Efficient perovskite solar cells via improved carrier management. Nature *590*, 587-593.
24. Jiang, Q., Tong, J. H., Xian, Y. M., Kerner, R. A., Dunfield, S. P., Xiao, C. X., Scheidt, R. A., Kuciauskas, D., Wang, X. M., Hautzinger, M. P., et al. (2022). Surface reaction for efficient and stable inverted perovskite solar cells. Nature *611,* 278-283.
25. Peng, W., Mao, K., Cai, F., Meng, H., Zhu, Z., Li, T., Yuan, S., Xu, Z., Feng, X., Xu, J., et al. (2023). Reducing nonradiative recombination in perovskite solar cells with a porous insulator contact. Science *379*, 683-690.
26. Jeong, J., Kim, M., Seo, J., Lu, H. Z., Ahlawat, P., Mishra, A., Yang, Y. G., Hope, M. A., Eickemeyer, F. T., Kim, M., et al. (2021). Pseudo-halide anion engineering for alpha-$FAPbI_3$ perovskite solar cells. Nature *592*, 381-385.
27. Kim, M., Jeong, J., Lu, H. Z., Lee, T. K., Eickemeyer, F. T., Liu, Y. H., Choi, I. W., Choi, S. J., Jo, Y., Kim, H. B., et al. (2022). Conformal quantum dot-SnO_2 layers as electron transporters for efficient perovskite solar cells. Science *375*, 302-306.
28. Min, H., Lee, D., Kim, J., Kim, G., Lee, K. S., Kim, J., Paik, M. J., Kim, Y. K., Kim, K. S., Kim, M. G., et al. (2021). Perovskite solar cells with atomically coherent interlayers on SnO_2 electrodes. Nature *598*, 444-450.
29. Tan, Q., Li, Z., Luo, G., Zhang, X., Che, B., Chen, G., Gao, H., He, D., Ma, G., Wang, J., et al. (2023). Inverted perovskite solar cells using dimethylacridine-based dopants. Nature *620*, 545-551.
30. Jeong, M. J., Moon, C. S., Lee, S., Im, J. M., Woo, M. Y., Lee, J. H., Cho, H., Jeon, S. W., Noh, J. H. (2023). Boosting radiation of stacked halide layer for perovskite solar cells with efficiency over 25%. Joule *7*, 112-127.
31. Park, J., Kim, J., Yun, H.-S., Paik, M. J., Noh, E., Mun, H. J., Kim, M. G., Shin, T. J., Seok, S. I. (2023). Controlled growth of perovskite layers with volatile alkylammonium chlorides. Nature *616*, 724-730.
32. Huang, Z. J., Bai, Y., Huang, X. D., Li, J. T., Wu, Y. T., Chen, Y. H., Li, K. L., Niu, X. X., Li, N. X., Liu, G. L., et al. (2023). Anion–π interactions suppress phase impurities in $FAPbI_3$ solar cells. Nature *623*, 531-537.